

Analyse des performances des requêtes OLAP

Trino - STACK B - TOM

Projet NutriScan - Couche Gold / hive.nutriscan_gold

Objectif du document

Ce document rassemble les requêtes OLAP exécutées avec EXPLAIN ANALYZE, les résultats de performance relevés dans Trino et les captures d'écran de l'onglet Query Details dans trino.

1. Méthodologie

Chaque requête OLAP a été exécutée avec EXPLAIN ANALYZE dans Trino. Cette commande exécute réellement la requête et produit un plan d'exécution détaillé avec les temps d'analyse, de planification et d'exécution, ainsi que les opérations principales réalisées par le moteur Trino.

Les requêtes ont été lancées une par une afin de limiter l'effet de la charge concurrente sur l'environnement Docker local, MinIO et Trino.

Requete 1 - ROLLUP categories et pays

Objectif : Calculer le Nutri-Score moyen par categorie et par pays, avec des sous-totaux grace a ROLLUP.

Type OLAP : ROLLUP / aggregation multidimensionnelle

Requête exécutée

```
EXPLAIN ANALYZE
SELECT
    dc.category_name,
    dco.country_name,
    ROUND(AVG(fn.nutriscore_score), 2) AS nutriscore_moyen,
    COUNT(DISTINCT fn.product_sk) AS nb_produits
FROM hive.nutriscan_gold.fact_nutrition fn
JOIN hive.nutriscan_gold.dim_category dc
    ON fn.category_sk = dc.category_sk
JOIN hive.nutriscan_gold.dim_country dco
    ON fn.country_sk = dco.country_sk
GROUP BY ROLLUP(dc.category_name, dco.country_name)
ORDER BY nutriscore_moyen DESC,
         dc.category_name NULLS LAST,
         dco.country_name NULLS LAST
LIMIT 10;
```

Résultats de performance relevés

Indicateur	Valeur
Trino version	435
Queued	14,72 ms
Analysis	2,15 s
Planning	6,31 s
Execution	1,21 min
Splits traités	35

Données traitées principales

- fact_nutrition scannee : 4 770 925 lignes
- dim_category scan : 35 563 lignes
- dim_country scan : 384 lignes
- Sortie finale : 10 lignes
- Lecture physique fact_nutrition : 25,06 MB

Emplacement capture Trino UI - Query Details

 QUERY DETAILS

VERSION
435

ENVIRONMENT
PRODUCTION

UPTIME
35.03m

Log Out

20260709_135838_00007_7kpwn

Overview

Live Plan

Stage Performance

Splits

JSON

FINISHED

Session

User	admin
Principal	admin
Source	Apache Superset
Catalog	hive
Schema	nutriscan_gold
Time zone	UTC
Client Address	172.18.0.8
Client Tags	
Session Properties	
Resource Estimates	

Execution

Resource Group	global
Submission Time	2026-07-09 3:58pm
Completion Time	2026-07-09 3:59pm
Elapsed Time	1.25m
Queued Time	14.72ms
Analysis Time	2.15s
Planning Time	6.31s
Execution Time	1.22m

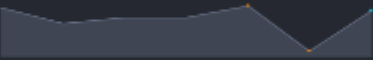
Resource Utilization Summary

CPU Time	26.71s
Planning CPU Time	1.22s
Scheduled Time	3.20m
Input Rows	4.81M
Input Data	91.9MB
Physical Input Rows	4.81M
Physical Input Data	26.0MB
Physical Input Read Time	855.00ms
Internal Network Rows	13.1M
Internal Network Data	341MB
Peak User Memory	624MB
Peak Total Memory	624MB
Cumulative User Memory	9.24GB*seconds
Output Rows	1.00
Output Data	14.5KB
Written Rows	0.00
Logical Written Data	0B
Physical Written Data	0B

Timeline

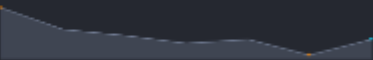
Parallelism

0.17



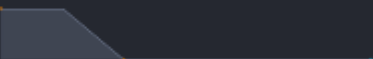
Scheduled Time/s

0.66



Input Rows/s

0.00



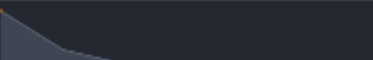
Input Bytes/s

0B



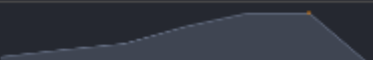
Physical Input Bytes/s

0B



Memory Utilization

0B



Résultat R1 - EXPLAIN ANALYZE

Trino version: 435 Queued: 14.72ms, Analysis: 2.15s, Planning: 6.31s, Execution: 1.21m

Fragment 1 [SINGLE]
CPU: 5.22ms, Scheduled: 28.91ms, Blocked 3.17m (Input: 2.13m, Output: 0.00ns), Input: 20 rows (1.04kB); per task: avg.: 20.00 std.dev.: 0.00, Output: 10 rows (526B)
Output layout: [round, country_name\$gid, count, category_name\$gid]
Output partitioning: SINGLE []

```

TopN[count = 10, orderBy = [round DESC NULLS LAST, category_name$gid ASC NULLS LAST, country_name$gid ASC NULLS LAST]]
  Layout: [round:double, country_name$gid:varchar, count:bigint, category_name$gid:varchar]
  Estimates: {rows: 10 (1.25kB), cpu: ?, memory: ?, network: ?}
  CPU: 2.00ms (0.01%), Scheduled: 20.00ms (0.01%), Blocked: 0.00ns (0.00%), Output: 10 rows (526B)
  Input avg.: 20.00 rows, Input std.dev.: 0.00%

LocalExchange[partitioning = SINGLE]
  Layout: [round:double, country_name$gid:varchar, count:bigint, category_name$gid:varchar]
  Estimates: {rows: ? (?), cpu: 0, memory: 0B, network: 0B}
  CPU: 0.00ns (0.00%), Scheduled: 1.00ms (0.00%), Blocked: 1.04m (7.57%), Output: 20 rows (1.04kB)
  Input avg.: 10.00 rows, Input std.dev.: 0.00%

RemoteSource[sourceFragmentIds = [2]] Layout: [round:double, country_name$gid:varchar, count:bigint, category_name$gid:varchar]
  CPU: 1.00ms (0.00%), Scheduled: 7.00ms (0.00%), Blocked: 2.13m (15.49%), Output: 20 rows (1.04kB)
  Input avg.: 10.00 rows, Input std.dev.: 0.00%

Fragment 2 [HASH]
  CPU: 583.60ms, Scheduled: 2.26s, Blocked 4.00m (Input: 2.01m, Output: 0.00ns), Input: 91892 rows (5.99MB); per task: avg.: 91892.00 std.dev.: 0.00,
  Output: 20 rows (1.04kB)
  Output layout: [round, country_name$gid, count, category_name$gid]
  Output partitioning: SINGLE []
TopNPartial[count = 10, orderBy = [round DESC NULLS LAST, category_name$gid ASC NULLS LAST, country_name$gid ASC NULLS LAST]]
  Layout: [round:double, country_name$gid:varchar, count:bigint, category_name$gid:varchar]
  CPU: 122.00ms (0.46%), Scheduled: 374.00ms (0.20%), Blocked: 0.00ns (0.00%), Output: 20 rows (1.04kB)
  Input avg.: 39668.50 rows, Input std.dev.: 0.04%

Project[]
  Layout: [round:double, country_name$gid:varchar, count:bigint, category_name$gid:varchar]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 45.00ms (0.17%), Scheduled: 203.00ms (0.11%), Blocked: 0.00ns (0.00%), Output: 79337 rows (3.76MB)
  Input avg.: 39668.50 rows, Input std.dev.: 0.04%
  round := round("avg", 2)

Project[]
  Layout: [category_name$gid:varchar, country_name$gid:varchar, avg:double, count:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 11.00ms (0.04%), Scheduled: 64.00ms (0.03%), Blocked: 0.00ns (0.00%), Output: 79337 rows (3.76MB)
  Input avg.: 39668.50 rows, Input std.dev.: 0.04%

Aggregate[type = FINAL, keys = [category_name$gid, country_name$gid, groupid]]
  Layout: [category_name$gid:varchar, country_name$gid:varchar, groupid:bigint, avg:double, count:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
  CPU: 345.00ms (1.30%), Scheduled: 1.36s (0.71%), Blocked: 0.00ns (0.00%), Output: 79337 rows (4.44MB)
  Input avg.: 45946.00 rows, Input std.dev.: 0.09%
  avg := avg("avg_8")
  count := count("count_9")

LocalExchange[partitioning = HASH, arguments = ["category_name$gid", "country_name$gid", "groupid"]]
  Layout: [category_name$gid:varchar, country_name$gid:varchar, groupid:bigint, avg_8:row(double, bigint), count_9:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 47.00ms (0.18%), Scheduled: 183.00ms (0.10%), Blocked: 2.00m (14.55%), Output: 91892 rows (5.99MB)
  Input avg.: 45946.00 rows, Input std.dev.: 26.88%

RemoteSource[sourceFragmentIds = [3]] Layout: [category_name$gid:varchar, country_name$gid:varchar, groupid:bigint, avg_8:row(double, bigint),
count_9:bigint]
  CPU: 10.00ms (0.04%), Scheduled: 61.00ms (0.03%), Blocked: 2.01m (14.62%), Output: 91892 rows (5.99MB)
  Input avg.: 45946.00 rows, Input std.dev.: 26.88%

Fragment 3 [HASH]
  CPU: 14.94s, Scheduled: 1.35m, Blocked 1.52m (Input: 42.27s, Output: 0.00ns), Input: 6168102 rows (281.12MB); per task: avg.: 6168102.00 std.dev.:
0.00, Output: 91892 rows (6.34MB)
  Output layout: [category_name$gid, country_name$gid, groupid, avg_8, count_9]
  Output partitioning: HASH [category_name$gid, country_name$gid, groupid]
Aggregate[type = PARTIAL, keys = [category_name$gid, country_name$gid, groupid]]
  Layout: [category_name$gid:varchar, country_name$gid:varchar, groupid:bigint, avg_8:row(double, bigint), count_9:bigint]
  CPU: 3.02s (11.41%), Scheduled: 17.20s (9.01%), Blocked: 0.00ns (0.00%), Output: 91892 rows (6.34MB)
  Input avg.: 3084051.00 rows, Input std.dev.: 0.01%
  avg_8 := avg("expr")
  count_9 := count("product_sk") (mask = product_sk$distinct)

MarkDistinct[distinct = [category_name$gid:varchar, country_name$gid:varchar, groupid:bigint, product_sk:integer], marker = product_sk$distinct]
  Layout: [category_name$gid:varchar, country_name$gid:varchar, product_sk:integer, expr:bigint, groupid:bigint, product_sk$distinct:boolean]
  CPU: 9.27s (34.99%), Scheduled: 42.11s (22.05%), Blocked: 0.00ns (0.00%), Output: 6168102 rows (292.88MB)

```

```

|
Input avg.: 3084051.00 rows, Input std.dev.: 0.01%
└─
LocalExchange[partitioning = HASH, arguments = ["category_name$gid", "country_name$gid", "groupid", "product_sk"]]
|
Layout: [category_name$gid:vchar, country_name$gid:vchar, product_sk:integer, expr:bigint, groupid:bigint]
|
Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
|
CPU: 2.07s (7.83%), Scheduled: 16.83s (8.81%), Blocked: 48.97s (5.94%), Output: 6168102 rows (281.12MB)
|
Input avg.: 3084051.00 rows, Input std.dev.: 4.74%
└─
RemoteSource[sourceFragmentIds = [4]] Layout: [category_name$gid:vchar, country_name$gid:vchar, product_sk:integer, expr:bigint, groupid:bigint]
CPU: 532.00ms (2.01%), Scheduled: 4.95s (2.59%), Blocked: 42.27s (5.12%), Output: 6168102 rows (281.12MB)
Input avg.: 3084051.00 rows, Input std.dev.: 4.74%

Fragment 4 [HASH]
CPU: 4.51s, Scheduled: 42.47s, Blocked 2.88m (Input: 1.23m, Output: 4.58s), Input: 2056418 rows (74.67MB); per task: avg.: 2056418.00 std.dev.: 0.00,
Output: 6168102 rows (281.12MB)
Output layout: [category_name$gid, country_name$gid, product_sk, expr, groupid]
Output partitioning: HASH [category_name$gid, country_name$gid, groupid, product_sk]
GroupId[symbols = [[]], [category_name], [category_name, country_name]]
|
Layout: [category_name$gid:vchar, country_name$gid:vchar, product_sk:integer, expr:bigint, groupid:bigint]
|
CPU: 3.12s (11.76%), Scheduled: 29.68s (15.54%), Blocked: 4.58s (0.56%), Output: 6168102 rows (281.12MB)
|
Input avg.: 1028017.00 rows, Input std.dev.: 10.44%
|
country_name$gid := country_name
|
category_name$gid := category_name
└─
Project[]
|
Layout: [category_name:vchar, product_sk:integer, country_name:vchar, expr:bigint]
|
Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
|
CPU: 295.00ms (1.11%), Scheduled: 2.88s (1.51%), Blocked: 0.00ns (0.00%), Output: 2056034 rows (98.61MB)
|
Input avg.: 1028017.00 rows, Input std.dev.: 10.44%
|
expr := CAST("nutriscore_score" AS bigint)
└─
InnerJoin[criteria = ("country_sk" = "country_sk_4"), distribution = PARTITIONED]
|
Layout: [product_sk:integer, nutriscore_score:integer, category_name:vchar, country_name:vchar]
|
Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
|
CPU: 977.00ms (3.69%), Scheduled: 8.26s (4.33%), Blocked: 27.62s (3.35%), Output: 2056034 rows (90.77MB)
|
Left (probe)
Input avg.: 1028017.00 rows, Input std.dev.: 10.44%
|
Right (build)
Input avg.: 192.00 rows, Input std.dev.: 1.04%
|
Distribution: PARTITIONED
|
dynamicFilterAssignments = {country_sk_4 -> #df_882}
└─
RemoteSource[sourceFragmentIds = [5]]
|
Layout: [product_sk:integer, country_sk:integer, nutriscore_score:integer, category_name:vchar]
|
CPU: 110.00ms (0.42%), Scheduled: 1.62s (0.85%), Blocked: 41.87s (5.08%), Output: 2056034 rows (74.67MB)
|
Input avg.: 1028017.00 rows, Input std.dev.: 10.44%
└─
LocalExchange[partitioning = HASH, arguments = ["country_sk_4"]]
|
Layout: [country_sk_4:integer, country_name:vchar]
|
Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
|
CPU: 1.00ms (0.00%), Scheduled: 1.00ms (0.00%), Blocked: 35.28s (4.28%), Output: 384 rows (7.63kB)
|
Input avg.: 192.00 rows, Input std.dev.: 100.00%
└─
RemoteSource[sourceFragmentIds = [8]] Layout: [country_sk_4:integer, country_name:vchar]
CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 32.01s (3.88%), Output: 384 rows (7.63kB)
Input avg.: 192.00 rows, Input std.dev.: 100.00%

Fragment 5 [HASH]
CPU: 3.09s, Scheduled: 26.97s, Blocked 2.68m (Input: 1.31m, Output: 0.00ns), Input: 4806488 rows (91.92MB); per task: avg.: 4806488.00 std.dev.: 0.00,
Output: 2056034 rows (74.67MB)
Output layout: [product_sk, country_sk, nutriscore_score, category_name]
Output partitioning: HASH [country_sk]
InnerJoin[criteria = ("category_sk" = "category_sk_0"), distribution = PARTITIONED]
|
Layout: [product_sk:integer, country_sk:integer, nutriscore_score:integer, category_name:vchar]
|
Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
|
CPU: 2.82s (10.66%), Scheduled: 24.86s (13.02%), Blocked: 45.41s (5.51%), Output: 2056034 rows (74.67MB)
|
Left (probe)
Input avg.: 2385462.50 rows, Input std.dev.: 2.47%
|
Right (build)
Input avg.: 17781.50 rows, Input std.dev.: 0.37%
|
Distribution: PARTITIONED
|
dynamicFilterAssignments = {category_sk_0 -> #df_883}
└─
RemoteSource[sourceFragmentIds = [6]]
|
Layout: [product_sk:integer, category_sk:integer, country_sk:integer, nutriscore_score:integer]

```

```

CPU: 192.00ms (0.72%), Scheduled: 1.64s (0.86%), Blocked: 41.12s (4.99%), Output: 4770925 rows (91.00MB)
└─
Input avg.: 2385462.50 rows, Input std.dev.: 2.47%
LocalExchange[partitioning = HASH, arguments = ["category_sk_0"]]
└─
Layout: [category_sk_0:integer, category_name:varchar]
└─
Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
└─
CPU: 64.00ms (0.24%), Scheduled: 390.00ms (0.20%), Blocked: 37.47s (4.54%), Output: 35563 rows (943.60kB)
└─
Input avg.: 17781.50 rows, Input std.dev.: 100.00%
RemoteSource[sourceFragmentIds = [7]] Layout: [category_sk_0:integer, category_name:varchar]
└─
CPU: 3.00ms (0.01%), Scheduled: 3.00ms (0.00%), Blocked: 37.39s (4.53%), Output: 35563 rows (943.60kB)
Input avg.: 17781.50 rows, Input std.dev.: 100.00%

Fragment 6 [SOURCE]
CPU: 3.07s, Scheduled: 31.30s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 4770925 rows (91.00MB); per task: avg.: 4770925.00 std.dev.: 0.00, Output: 4770925 rows (91.00MB)
Output layout: [product_sk, category_sk, country_sk, nutriscore_score]
Output partitioning: HASH [category_sk]
ScanFilter[table = hive:nutriscan_gold:fact_nutrition, dynamicFilters = {"country_sk" = #df_882, "category_sk" = #df_883}] Layout: [product_sk:integer, category_sk:integer, country_sk:integer, nutriscore_score:integer]
└─
Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
CPU: 3.07s (11.59%), Scheduled: 31.25s (16.36%), Blocked: 0.00ns (0.00%), Output: 4770925 rows (91.00MB)
Input avg.: 681560.71 rows, Input std.dev.: 244.95% category_sk := category_sk:int:REGULAR product_sk := product_sk:int:REGULAR nutriscore_score := nutriscore_score:int:REGULAR country_sk := country_sk:int:REGULAR Input: 4770925 rows (91.00MB), Filtered: 0.00%, Physical input: 25.06MB, Physical input time: 534.00ms
Dynamic filters:
- df_882, [ SortedRangeSet[type=integer, ranges=1, {[1,384]}] ], collection time=22.45s
- df_883, ALL, collection time=23.54s

Fragment 7 [SOURCE]
CPU: 194.21ms, Scheduled: 3.75s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 35563 rows (943.60kB); per task: avg.: 35563.00 std.dev.: 0.00, Output: 35563 rows (943.60kB)
Output layout: [category_sk_0, category_name]
Output partitioning: HASH [category_sk_0]
TableScan[table = hive:nutriscan_gold:dim_category] Layout: [category_sk_0:integer, category_name:varchar]
└─
Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
CPU: 194.00ms (0.73%), Scheduled: 3.75s (1.97%), Blocked: 0.00ns (0.00%), Output: 35563 rows (943.60kB)
Input avg.: 35563.00 rows, Input std.dev.: 0.00% category_name := category_name:string:REGULAR category_sk_0 := category_sk:int:REGULAR Input: 35563 rows (943.60kB), Physical input: 964.47kB, Physical input time: 167.44ms

Fragment 8 [SOURCE]
CPU: 164.01ms, Scheduled: 3.25s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 384 rows (7.63kB); per task: avg.: 384.00 std.dev.: 0.00, Output: 384 rows (7.63kB)
Output layout: [country_sk_4, country_name]
Output partitioning: HASH [country_sk_4]
TableScan[table = hive:nutriscan_gold:dim_country] Layout: [country_sk_4:integer, country_name:varchar]
└─
Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
CPU: 163.00ms (0.62%), Scheduled: 3.25s (1.70%), Blocked: 0.00ns (0.00%), Output: 384 rows (7.63kB)
Input avg.: 384.00 rows, Input std.dev.: 0.00% country_sk_4 := country_sk:int:REGULAR country_name := country_name:string:REGULAR Input: 384 rows (7.63kB), Physical input: 10.63kB, Physical input time: 154.21ms

```

Requete 2 - RANK des marques par Nutri-Score

Objectif : Classer les marques dans chaque categorie selon le Nutri-Score moyen et conserver les trois meilleures marques par categorie.

Type OLAP : RANK / fonction de fenetrage

Requête exécutée

```
EXPLAIN ANALYZE
WITH ranked AS (
  SELECT
    db.brand_name,
    dc.category_name,
    ROUND(AVG(fn.nutriscore_score), 2) AS nutriscore_moyen,
    RANK() OVER (
      PARTITION BY dc.category_name
      ORDER BY AVG(fn.nutriscore_score) ASC
    ) AS rang
  FROM hive.nutriscan_gold.fact_nutrition fn
  JOIN hive.nutriscan_gold.dim_brand db
    ON fn.brand_sk = db.brand_sk
  JOIN hive.nutriscan_gold.dim_category dc
    ON fn.category_sk = dc.category_sk
  GROUP BY db.brand_name, dc.category_name
)
SELECT *
FROM ranked
WHERE rang <= 3
ORDER BY nutriscore_moyen ASC, category_name, rang;
```

Résultats de performance relevés

Indicateur	Valeur
Trino version	435
Queued	1,82 ms
Analysis	788,40 ms
Planning	918,33 ms
Execution	15,54 s
Splits traités	33

Données traitées principales

- fact_nutrition scannee : 4 770 925 lignes
- dim_brand scan : 348 775 lignes
- dim_category scan : 35 563 lignes
- Sortie finale : 35 694 lignes
- Lecture physique fact_nutrition : 10,68 MB

 QUERY DETAILS

VERSION
435

ENVIRONMENT
PRODUCTION

UPTIME
36.15m

Log Out

20260709_140030_00018_7kpwn

Overview

Live Plan

Stage Performance

Splits

JSON

FINISHED

Session

User	admin
Principal	admin
Source	Apache Superset
Catalog	hive
Schema	nutriscan_gold
Time zone	UTC
Client Address	172.18.0.8
Client Tags	
Session Properties	
Resource Estimates	

Execution

Resource Group	global
Submission Time	2026-07-09 4:00pm
Completion Time	2026-07-09 4:00pm
Elapsed Time	16.45s
Queued Time	1.82ms
Analysis Time	788.40ms
Planning Time	918.33ms
Execution Time	15.56s

Resource Utilization Summary

CPU Time	11.81s
Planning CPU Time	350.32ms
Scheduled Time	56.29s
Input Rows	5.16M
Input Data	76.9MB
Physical Input Rows	5.16M
Physical Input Data	16.1MB
Physical Input Read Time	412.00ms
Internal Network Rows	8.66M
Internal Network Data	108MB
Peak User Memory	76.0MB
Peak Total Memory	76.0MB
Cumulative User Memory	631MB*seconds
Output Rows	1.00
Output Data	12.6KB
Written Rows	0.00
Logical Written Data	0B
Physical Written Data	0B

Timeline

Parallelism	0.72
Scheduled Time/s	3.42
Input Rows/s	313K
Input Bytes/s	4.67MB
Physical Input Bytes/s	39.1MB
Memory Utilization	0B

Résultat R2 - EXPLAIN ANALYZE

```
Trino version: 435 Queued: 1.82ms, Analysis: 788.40ms, Planning: 918.33ms, Execution: 15.54s

Fragment 1 [ROUND_ROBIN]
  CPU: 545.99ms, Scheduled: 1.03s, Blocked 39.67s (Input: 26.00s, Output: 0.00ns), Input: 35694 rows (1.85MB); per task: avg.: 35694.00 std.dev.: 0.00,
  Output: 35694 rows (1.85MB)
  Output layout: [round, brand_name, category_name, rank]
  Output partitioning: SINGLE []
  LocalMerge[orderBy = [round ASC NULLS LAST, category_name ASC NULLS LAST, rank ASC NULLS LAST]]
```



```

    | Layout: [round:double, brand_name:varchar, category_name:varchar, rank:bigint]
  Estimates: {rows: ? (?), cpu: 0, memory: 0B, network: 0B}
  CPU: 108.00ms (0.92%), Scheduled: 251.00ms (0.45%), Blocked: 13.68s (7.97%), Output: 35694 rows (1.85MB)
  Input avg.: 17847.00 rows, Input std.dev.: 55.97%
  |
PartialSort[orderBy = [round ASC NULLS LAST, category_name ASC NULLS LAST, rank ASC NULLS LAST]]
  | Layout: [round:double, brand_name:varchar, category_name:varchar, rank:bigint]
  CPU: 428.00ms (3.66%), Scheduled: 745.00ms (1.33%), Blocked: 0.00ns (0.00%), Output: 35694 rows (1.78MB)
  Input avg.: 17847.00 rows, Input std.dev.: 55.97%
  |
RemoteSource[sourceFragmentIds = [2]] Layout: [round:double, brand_name:varchar, category_name:varchar, rank:bigint]
  CPU: 8.00ms (0.07%), Scheduled: 35.00ms (0.06%), Blocked: 26.00s (15.15%), Output: 35694 rows (1.85MB)
  Input avg.: 17847.00 rows, Input std.dev.: 55.97%

Fragment 2 [HASH]
  CPU: 254.51ms, Scheduled: 501.13ms, Blocked 50.41s (Input: 25.44s, Output: 0.00ns), Input: 38537 rows (1.65MB); per task: avg.: 38537.00 std.dev.: 0.00, Output: 35694 rows (1.85MB)
  Output layout: [round, brand_name, category_name, rank]
  Output partitioning: ROUND_ROBIN []
Project[]
  | Layout: [round:double, brand_name:varchar, category_name:varchar, rank:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 15.00ms (0.13%), Scheduled: 25.00ms (0.04%), Blocked: 0.00ns (0.00%), Output: 35694 rows (1.85MB)
  Input avg.: 17847.00 rows, Input std.dev.: 1.41%
  | round := round("avg", 2)
  |
TopNRanking[partitionBy = [category_name], orderBy = [avg ASC NULLS LAST], limit = 3]
  | Layout: [brand_name:varchar, category_name:varchar, avg:double, rank:bigint]
  CPU: 219.00ms (1.87%), Scheduled: 414.00ms (0.74%), Blocked: 0.00ns (0.00%), Output: 35694 rows (1.85MB)
  Input avg.: 19268.50 rows, Input std.dev.: 1.61%
  | rank := RANK
  |
LocalExchange[partitioning = HASH, arguments = ["category_name"]]
  | Layout: [brand_name:varchar, category_name:varchar, avg:double]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 17.00ms (0.15%), Scheduled: 43.00ms (0.08%), Blocked: 24.97s (14.55%), Output: 38537 rows (1.65MB)
  Input avg.: 19268.50 rows, Input std.dev.: 0.80%
  |
RemoteSource[sourceFragmentIds = [3]] Layout: [brand_name:varchar, category_name:varchar, avg:double]
  CPU: 2.00ms (0.02%), Scheduled: 2.00ms (0.00%), Blocked: 25.44s (14.82%), Output: 38537 rows (1.65MB)
  Input avg.: 19268.50 rows, Input std.dev.: 0.80%

Fragment 3 [HASH]
  CPU: 1.62s, Scheduled: 6.83s, Blocked 38.20s (Input: 20.34s, Output: 0.00ns), Input: 315924 rows (17.54MB); per task: avg.: 315924.00 std.dev.: 0.00, Output: 38537 rows (1.65MB)
  Output layout: [brand_name, category_name, avg]
  Output partitioning: HASH [category_name]
TopNRanking[partitionBy = [category_name], orderBy = [avg ASC NULLS LAST], limit = 3]
  | Layout: [brand_name:varchar, category_name:varchar, avg:double]
  CPU: 796.00ms (6.81%), Scheduled: 1.93s (3.44%), Blocked: 0.00ns (0.00%), Output: 38537 rows (1.65MB)
  Input avg.: 141461.00 rows, Input std.dev.: 0.13%
  | rank := RANK
  |
Aggregate[type = FINAL, keys = [brand_name, category_name]]
  | Layout: [brand_name:varchar, category_name:varchar, avg:double]
  Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
  CPU: 664.00ms (5.68%), Scheduled: 3.96s (7.06%), Blocked: 0.00ns (0.00%), Output: 282922 rows (13.06MB)
  Input avg.: 157962.00 rows, Input std.dev.: 0.15%
  | avg := avg("avg_9")
  |
LocalExchange[partitioning = HASH, arguments = ["brand_name", "category_name"]]
  | Layout: [brand_name:varchar, category_name:varchar, avg_9:row(double, bigint)]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 133.00ms (1.14%), Scheduled: 788.00ms (1.41%), Blocked: 17.96s (10.46%), Output: 315924 rows (17.54MB)
  Input avg.: 157962.00 rows, Input std.dev.: 3.72%
  |
RemoteSource[sourceFragmentIds = [4]] Layout: [brand_name:varchar, category_name:varchar, avg_9:row(double, bigint)]
  CPU: 22.00ms (0.19%), Scheduled: 146.00ms (0.26%), Blocked: 20.34s (11.85%), Output: 315924 rows (17.54MB)
  Input avg.: 157962.00 rows, Input std.dev.: 3.72%

Fragment 4 [HASH]
  CPU: 4.05s, Scheduled: 20.03s, Blocked 21.33s (Input: 1.11s, Output: 0.00ns), Input: 3115176 rows (73.46MB); per task: avg.: 3115176.00 std.dev.: 0.00, Output: 315924 rows (18.75MB)
  Output layout: [brand_name, category_name, avg_9]
  Output partitioning: HASH [brand_name, category_name]

```

```

Aggregate[type = PARTIAL, keys = [brand_name, category_name]]
| Layout: [brand_name:vvarchar, category_name:vvarchar, avg_9:row(double, bigint)]
| CPU: 2.94s (25.18%), Scheduled: 14.03s (25.04%), Blocked: 0.00ns (0.00%), Output: 315924 rows (18.75MB)
| Input avg.: 805322.50 rows, Input std.dev.: 13.32%
| avg_9 := avg("expr")
Project[]
| Layout: [brand_name:vvarchar, category_name:vvarchar, expr:bigint]
| Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
| CPU: 353.00ms (3.02%), Scheduled: 1.41s (2.52%), Blocked: 0.00ns (0.00%), Output: 1610645 rows (78.65MB)
| Input avg.: 805322.50 rows, Input std.dev.: 13.32%
| expr := CAST("nutriscore_score" AS bigint)
InnerJoin[criteria = ("category_sk" = "category_sk_4"), distribution = PARTITIONED]
| Layout: [nutriscore_score:integer, brand_name:vvarchar, category_name:vvarchar]
| Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
| CPU: 647.00ms (5.54%), Scheduled: 4.19s (7.48%), Blocked: 20.04s (11.67%), Output: 1610645 rows (78.50MB)
| Left (probe)
| Input avg.: 1539806.50 rows, Input std.dev.: 12.00%
| Right (build)
| Input avg.: 17781.50 rows, Input std.dev.: 0.37%
| Distribution: PARTITIONED
| dynamicFilterAssignments = {category_sk_4 -> #df_843}
RemoteSource[sourceFragmentIds = [5]]
| Layout: [category_sk:integer, nutriscore_score:integer, brand_name:vvarchar]
| CPU: 107.00ms (0.92%), Scheduled: 357.00ms (0.64%), Blocked: 1.06s (0.62%), Output: 3079613 rows (72.54MB)
| Input avg.: 1539806.50 rows, Input std.dev.: 12.00%
LocalExchange[partitioning = HASH, arguments = ["category_sk_4"]]
| Layout: [category_sk_4:integer, category_name:vvarchar]
| Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
| CPU: 3.00ms (0.03%), Scheduled: 20.00ms (0.04%), Blocked: 102.00ms (0.06%), Output: 35563 rows (943.60kB)
| Input avg.: 17781.50 rows, Input std.dev.: 100.00%
RemoteSource[sourceFragmentIds = [8]] Layout: [category_sk_4:integer, category_name:vvarchar]
| CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 52.00ms (0.03%), Output: 35563 rows (943.60kB)
| Input avg.: 17781.50 rows, Input std.dev.: 100.00%
Fragment 5 [HASH]
| CPU: 3.07s, Scheduled: 17.32s, Blocked 21.81s (Input: 3.38s, Output: 0.00ns), Input: 5119700 rows (75.94MB); per task: avg.: 5119700.00 std.dev.: 0.00, Output: 3079613 rows (72.53MB)
| Output layout: [category_sk, nutriscore_score, brand_name]
| Output partitioning: HASH [category_sk]
InnerJoin[criteria = ("brand_sk" = "brand_sk_0"), distribution = PARTITIONED]
| Layout: [category_sk:integer, nutriscore_score:integer, brand_name:vvarchar]
| Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
| CPU: 2.85s (24.38%), Scheduled: 15.66s (27.95%), Blocked: 17.05s (9.93%), Output: 3079613 rows (72.53MB)
| Left (probe)
| Input avg.: 2385462.50 rows, Input std.dev.: 3.02%
| Right (build)
| Input avg.: 174387.50 rows, Input std.dev.: 0.21%
| Distribution: PARTITIONED
| dynamicFilterAssignments = {brand_sk_0 -> #df_844}
RemoteSource[sourceFragmentIds = [6]]
| Layout: [category_sk:integer, brand_sk:integer, nutriscore_score:integer]
| CPU: 161.00ms (1.38%), Scheduled: 1.35s (2.41%), Blocked: 1.99s (1.16%), Output: 4770925 rows (68.25MB)
| Input avg.: 2385462.50 rows, Input std.dev.: 3.02%
LocalExchange[partitioning = HASH, arguments = ["brand_sk_0"]]
| Layout: [brand_sk_0:integer, brand_name:vvarchar]
| Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
| CPU: 45.00ms (0.39%), Scheduled: 219.00ms (0.39%), Blocked: 1.61s (0.94%), Output: 348775 rows (7.69MB)
| Input avg.: 174387.50 rows, Input std.dev.: 6.05%
RemoteSource[sourceFragmentIds = [7]] Layout: [brand_sk_0:integer, brand_name:vvarchar]
| CPU: 8.00ms (0.07%), Scheduled: 76.00ms (0.14%), Blocked: 1.38s (0.80%), Output: 348775 rows (7.69MB)
| Input avg.: 174387.50 rows, Input std.dev.: 6.05%
Fragment 6 [SOURCE]
| CPU: 1.86s, Scheduled: 9.24s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 4770925 rows (68.25MB); per task: avg.: 4770925.00 std.dev.: 0.00, Output: 4770925 rows (68.25MB)
| Output layout: [category_sk, brand_sk, nutriscore_score]
| Output partitioning: HASH [brand_sk]

```

```
ScanFilter[table = hive:nutriscan_gold:fact_nutrition, dynamicFilters = {"category_sk" = #df_843, "brand_sk" = #df_844}] Layout: [category_sk:integer, brand_sk:integer, nutriscore_score:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 1.86s (15.95%), Scheduled: 9.24s (16.48%), Blocked: 0.00ns (0.00%), Output: 4770925 rows (68.25MB)
  Input avg.: 681560.71 rows, Input std.dev.: 244.95% category_sk := category_sk:int:REGULAR nutriscore_score := nutriscore_score:int:REGULAR brand_sk := brand_sk:int:REGULAR Input: 4770925 rows (68.25MB), Filtered: 0.00%, Physical input: 10.68MB, Physical input time: 143.87ms
Dynamic filters:
- df_843, ALL, collection time=1.36s
- df_844, ALL, collection time=1.26s
```

Fragment 7 [SOURCE]

```
  CPU: 253.96ms, Scheduled: 689.89ms, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 348775 rows (7.69MB); per task: avg.: 348775.00 std.dev.: 0.00, Output: 348775 rows (7.69MB)
  Output layout: [brand_sk_0, brand_name]
  Output partitioning: HASH [brand_sk_0]
  TableScan[table = hive:nutriscan_gold:dim_brand] Layout: [brand_sk_0:integer, brand_name:varchar]
    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    CPU: 254.00ms (2.17%), Scheduled: 690.00ms (1.23%), Blocked: 0.00ns (0.00%), Output: 348775 rows (7.69MB)
    Input avg.: 348775.00 rows, Input std.dev.: 0.00% brand_sk_0 := brand_sk:int:REGULAR brand_name := brand_name:string:REGULAR Input: 348775 rows (7.69MB), Physical input: 4.48MB, Physical input time: 83.23ms
```

Fragment 8 [SOURCE]

```
  CPU: 41.59ms, Scheduled: 454.93ms, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 35563 rows (943.60kB); per task: avg.: 35563.00 std.dev.: 0.00, Output: 35563 rows (943.60kB)
  Output layout: [category_sk_4, category_name]
  Output partitioning: HASH [category_sk_4]
  TableScan[table = hive:nutriscan_gold:dim_category] Layout: [category_sk_4:integer, category_name:varchar]
    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    CPU: 41.00ms (0.35%), Scheduled: 455.00ms (0.81%), Blocked: 0.00ns (0.00%), Output: 35563 rows (943.60kB)
    Input avg.: 35563.00 rows, Input std.dev.: 0.00% category_name := category_name:string:REGULAR category_sk_4 := category_sk:int:REGULAR Input: 35563 rows (943.60kB), Physical input: 964.47kB, Physical input time: 185.12ms
```

Requete 3 - Cumul temporel annuel

Objectif : Calculer le nombre de produits distincts par mois, puis un cumul progressif par annee.

Type OLAP : Fenetre cumulative / analyse temporelle

Requête exécutée

```
EXPLAIN ANALYZE
SELECT
  dd.year,
  dd.month,
  COUNT(DISTINCT fn.product_sk) AS nb_produits_mois,
  SUM(COUNT(DISTINCT fn.product_sk)) OVER (
    PARTITION BY dd.year
    ORDER BY dd.month
    ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
  ) AS nb_produits_cumule
FROM hive.nutriscan_gold.fact_nutrition fn
JOIN hive.nutriscan_gold.dim_date dd
  ON fn.date_sk = dd.date_sk
GROUP BY dd.year, dd.month
ORDER BY dd.year, dd.month;
```

Résultats de performance relevés

Indicateur	Valeur
Trino version	435
Queued	6,87 ms
Analysis	577,12 ms
Planning	1,76 s
Execution	20,50 s
Splits traités	30

Données traitées principales

- fact_nutrition scannee : 4 770 925 lignes
- dim_date scan : 3 856 lignes
- Sortie finale : 146 lignes
- Lecture physique fact_nutrition : 24,24 MB

Emplacement capture Trino UI - Query Details

 **QUERY DETAILS**

VERSION
435

ENVIRONMENT
PRODUCTION

UPTIME
37.38m

Log Out

20260709_140213_00019_7kpwn

Overview

Live Plan

Stage Performance

Splits

JSON

FINISHED

Session

User	admin
Principal	admin
Source	Apache Superset
Catalog	hive
Schema	nutriscan_gold
Time zone	UTC
Client Address	172.18.0.8
Client Tags	
Session Properties	
Resource Estimates	

Execution

Resource Group	global
Submission Time	2026-07-09 4:02pm
Completion Time	2026-07-09 4:02pm
Elapsed Time	21.21s
Queued Time	6.87ms
Analysis Time	577.12ms
Planning Time	1.76s
Execution Time	20.62s

Resource Utilization Summary

CPU Time	12.59s
Planning CPU Time	288.43ms
Scheduled Time	55.00s
Input Rows	4.77M
Input Data	45.6MB
Physical Input Rows	4.77M
Physical Input Data	24.3MB
Physical Input Read Time	269.00ms
Internal Network Rows	9.43M
Internal Network Data	89.7MB
Peak User Memory	197MB
Peak Total Memory	197MB
Cumulative User Memory	1.87GB*seconds
Output Rows	1.00
Output Data	9.71KB
Written Rows	0.00
Logical Written Data	0B
Physical Written Data	0B

Timeline

Parallelism	1.26
Scheduled Time/s	2.99
Input Rows/s	0.00
Input Bytes/s	0B
Physical Input Bytes/s	0B
Memory Utilization	0B

Résultat R3- EXPLAIN ANALYZE

Trino version: 435 Queued: 6.87ms, Analysis: 577.12ms, Planning: 1.76s, Execution: 20.50s

Fragment 1 [ROUND_ROBIN]
CPU: 7.77ms, Scheduled: 40.33ms, Blocked 54.54s (Input: 36.38s, Output: 0.00ns), Input: 146 rows (3.99kB); per task: avg.: 146.00 std.dev.: 0.00, Output: 146 rows (3.99kB)
Output layout: [year, month, count, sum]
Output partitioning: SINGLE []

```

LocalMerge[orderBy = [year ASC NULLS LAST, month ASC NULLS LAST]]
  Layout: [year:integer, month:integer, count:bigint, sum:bigint]
  Estimates: {rows: ? (?), cpu: 0, memory: 0B, network: 0B}
  CPU: 3.00ms (0.02%), Scheduled: 32.00ms (0.06%), Blocked: 18.17s (7.87%), Output: 146 rows (3.99kB)
  Input avg.: 73.00 rows, Input std.dev.: 17.81%

PartialSort[orderBy = [year ASC NULLS LAST, month ASC NULLS LAST]]
  Layout: [year:integer, month:integer, count:bigint, sum:bigint]
  CPU: 2.00ms (0.02%), Scheduled: 2.00ms (0.00%), Blocked: 0.00ns (0.00%), Output: 146 rows (3.99kB)
  Input avg.: 73.00 rows, Input std.dev.: 17.81%

RemoteSource[sourceFragmentIds = [2]] Layout: [year:integer, month:integer, count:bigint, sum:bigint]
  CPU: 0.00ns (0.00%), Scheduled: 5.00ms (0.01%), Blocked: 36.38s (15.75%), Output: 146 rows (3.99kB)
  Input avg.: 73.00 rows, Input std.dev.: 17.81%

Fragment 2 [HASH]
  CPU: 50.48ms, Scheduled: 290.01ms, Blocked 1.15m (Input: 34.86s, Output: 0.00ns), Input: 146 rows (2.71kB); per task: avg.: 146.00 std.dev.: 0.00,
  Output: 146 rows (3.99kB)
  Output layout: [year, month, count, sum]
  Output partitioning: ROUND_ROBIN []
  Window[partitionBy = [year], orderBy = [month ASC NULLS LAST]]
    Layout: [year:integer, month:integer, count:bigint, sum:bigint]
    CPU: 33.00ms (0.26%), Scheduled: 167.00ms (0.31%), Blocked: 0.00ns (0.00%), Output: 146 rows (3.99kB)
    Input avg.: 73.00 rows, Input std.dev.: 17.81%
    sum := sum("count") ROWS UNBOUNDED_PRECEDING CURRENT_ROW

LocalExchange[partitioning = HASH, arguments = ["year"]]
  Layout: [year:integer, month:integer, count:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 33.88s (14.67%), Output: 146 rows (2.71kB)
  Input avg.: 73.00 rows, Input std.dev.: 13.70%

RemoteSource[sourceFragmentIds = [3]] Layout: [year:integer, month:integer, count:bigint]
  CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 34.86s (15.09%), Output: 146 rows (2.71kB)
  Input avg.: 73.00 rows, Input std.dev.: 13.70%

Fragment 3 [HASH]
  CPU: 5.94ms, Scheduled: 38.89ms, Blocked 1.20m (Input: 36.05s, Output: 0.00ns), Input: 286 rows (5.31kB); per task: avg.: 286.00 std.dev.: 0.00,
  Output: 146 rows (2.71kB)
  Output layout: [year, month, count]
  Output partitioning: HASH [year]
  Aggregate[type = FINAL, keys = [year, month]]
    Layout: [year:integer, month:integer, count:bigint]
    Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
    CPU: 3.00ms (0.02%), Scheduled: 34.00ms (0.06%), Blocked: 0.00ns (0.00%), Output: 146 rows (2.71kB)
    Input avg.: 143.00 rows, Input std.dev.: 13.29%
    count := count("count_4")

LocalExchange[partitioning = HASH, arguments = ["year", "month"]]
  Layout: [year:integer, month:integer, count_4:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 1.00ms (0.01%), Scheduled: 2.00ms (0.00%), Blocked: 36.00s (15.58%), Output: 286 rows (5.31kB)
  Input avg.: 143.00 rows, Input std.dev.: 0.00%

RemoteSource[sourceFragmentIds = [4]] Layout: [year:integer, month:integer, count_4:bigint]
  CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 36.05s (15.61%), Output: 286 rows (5.31kB)
  Input avg.: 143.00 rows, Input std.dev.: 0.00%

Fragment 4 [HASH]
  CPU: 7.02s, Scheduled: 26.17s, Blocked 33.11s (Input: 20.16s, Output: 0.00ns), Input: 4651084 rows (66.53MB); per task: avg.: 4651084.00 std.dev.: 0.00,
  Output: 286 rows (5.31kB)
  Output layout: [year, month, count_4]
  Output partitioning: HASH [year, month]
  Aggregate[type = PARTIAL, keys = [year, month]]
    Layout: [year:integer, month:integer, count_4:bigint]
    CPU: 417.00ms (3.33%), Scheduled: 1.12s (2.05%), Blocked: 0.00ns (0.00%), Output: 286 rows (5.31kB)
    Input avg.: 2249361.00 rows, Input std.dev.: 0.05%
    count_4 := count("product_sk")

Aggregate[type = FINAL, keys = [year, month, product_sk]]
  Layout: [year:integer, month:integer, product_sk:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
  CPU: 5.84s (46.69%), Scheduled: 20.45s (37.47%), Blocked: 0.00ns (0.00%), Output: 4498722 rows (64.35MB)
  Input avg.: 2325542.00 rows, Input std.dev.: 0.06%

```

```

LocalExchange[partitioning = HASH, arguments = ["year", "month", "product_sk"]]
  Layout: [year:integer, month:integer, product_sk:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 668.00ms (5.34%), Scheduled: 4.10s (7.51%), Blocked: 13.10s (5.67%), Output: 4651084 rows (66.53MB)
  Input avg.: 2325542.00 rows, Input std.dev.: 1.32%

RemoteSource[sourceFragmentIds = [5]] Layout: [year:integer, month:integer, product_sk:integer]
  CPU: 75.00ms (0.60%), Scheduled: 368.00ms (0.67%), Blocked: 20.16s (8.73%), Output: 4651084 rows (66.53MB)
  Input avg.: 2325542.00 rows, Input std.dev.: 1.32%

Fragment 5 [HASH]
  CPU: 4.83s, Scheduled: 23.76s, Blocked 25.35s (Input: 1.51s, Output: 0.00ns), Input: 4774781 rows (45.55MB); per task: avg.: 4774781.00 std.dev.: 0.00, Output: 4651084 rows (66.53MB)
  Output layout: [year, month, product_sk]
  Output partitioning: HASH [year, month, product_sk]
  Aggregate[type = PARTIAL, keys = [year, month, product_sk]]
    Layout: [year:integer, month:integer, product_sk:integer]
    CPU: 3.40s (27.19%), Scheduled: 15.82s (28.99%), Blocked: 0.00ns (0.00%), Output: 4651084 rows (66.53MB)
    Input avg.: 2385462.50 rows, Input std.dev.: 16.21%

InnerJoin[criteria = ("date_sk" = "date_sk_0"), distribution = PARTITIONED]
  Layout: [product_sk:integer, year:integer, month:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
  CPU: 1.36s (10.84%), Scheduled: 7.74s (14.18%), Blocked: 292.00ms (0.13%), Output: 4770925 rows (68.25MB)
  Left (probe)
    Input avg.: 2385462.50 rows, Input std.dev.: 16.21%
  Right (build)
    Input avg.: 1928.00 rows, Input std.dev.: 1.92%
    Distribution: PARTITIONED
    maySkipOutputDuplicates = true
  dynamicFilterAssignments = {date_sk_0 -> #df_579}

RemoteSource[sourceFragmentIds = [6]]
  Layout: [product_sk:integer, date_sk:integer]
  CPU: 59.00ms (0.47%), Scheduled: 192.00ms (0.35%), Blocked: 1.03s (0.45%), Output: 4770925 rows (45.50MB)
  Input avg.: 2385462.50 rows, Input std.dev.: 16.21%

LocalExchange[partitioning = HASH, arguments = ["date_sk_0"]]
  Layout: [date_sk_0:integer, year:integer, month:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 1.00ms (0.01%), Scheduled: 4.00ms (0.01%), Blocked: 594.00ms (0.26%), Output: 3856 rows (56.48kB)
  Input avg.: 1928.00 rows, Input std.dev.: 100.00%

RemoteSource[sourceFragmentIds = [7]] Layout: [date_sk_0:integer, year:integer, month:integer]
  CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 481.00ms (0.21%), Output: 3856 rows (56.48kB)
  Input avg.: 1928.00 rows, Input std.dev.: 100.00%

Fragment 6 [SOURCE]
  CPU: 620.38ms, Scheduled: 4.38s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 4770925 rows (45.50MB); per task: avg.: 4770925.00 std.dev.: 0.00, Output: 4770925 rows (45.50MB)
  Output layout: [product_sk, date_sk]
  Output partitioning: HASH [date_sk]
  ScanFilter[table = hive:nutriscan_gold:fact_nutrition, dynamicFilters = {"date_sk" = #df_579}] Layout: [product_sk:integer, date_sk:integer]
    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    CPU: 619.00ms (4.95%), Scheduled: 4.39s (8.05%), Blocked: 0.00ns (0.00%), Output: 4770925 rows (45.50MB)
    Input avg.: 681560.71 rows, Input std.dev.: 244.95% product_sk := product_sk:int:REGULAR date_sk := date_sk:int:REGULAR Input: 4770925 rows (45.50MB), Filtered: 0.00%, Physical input: 24.24MB, Physical input time: 241.11ms
  Dynamic filters:
    - df_579, ALL, collection time=847.20ms

Fragment 7 [SOURCE]
  CPU: 30.20ms, Scheduled: 152.46ms, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 3856 rows (56.48kB); per task: avg.: 3856.00 std.dev.: 0.00, Output: 3856 rows (56.48kB)
  Output layout: [date_sk_0, year, month]
  Output partitioning: HASH [date_sk_0]
  TableScan[table = hive:nutriscan_gold:dim_date] Layout: [date_sk_0:integer, year:integer, month:integer]
    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    CPU: 31.00ms (0.25%), Scheduled: 152.00ms (0.28%), Blocked: 0.00ns (0.00%), Output: 3856 rows (56.48kB)
    Input avg.: 3856.00 rows, Input std.dev.: 0.00% date_sk_0 := date_sk:int:REGULAR month := month:int:REGULAR year := year:int:REGULAR Input: 3856 rows (56.48kB), Physical input: 32.92kB, Physical input time: 27.75ms

```


Requete 4 - PIVOT Nutri-Score par categorie

Objectif : Produire un tableau croise avec le nombre de produits par grade Nutri-Score a, b, c, d et e pour chaque categorie.

Type OLAP : PIVOT avec CASE WHEN

Requête exécutée

```
EXPLAIN ANALYZE
WITH produits_categories AS (
  SELECT DISTINCT
    fn.product_sk,
    fn.category_sk,
    LOWER(dp.nutriscore_grade) AS nutriscore_grade
  FROM hive.nutriscan_gold.fact_nutrition fn
  JOIN hive.nutriscan_gold.dim_product dp
    ON fn.product_sk = dp.product_sk
  WHERE dp.nutriscore_grade IS NOT NULL
)
SELECT
  dc.category_name,
  COUNT(DISTINCT CASE WHEN pc.nutriscore_grade = 'a' THEN pc.product_sk END) AS nb_grade_a,
  COUNT(DISTINCT CASE WHEN pc.nutriscore_grade = 'b' THEN pc.product_sk END) AS nb_grade_b,
  COUNT(DISTINCT CASE WHEN pc.nutriscore_grade = 'c' THEN pc.product_sk END) AS nb_grade_c,
  COUNT(DISTINCT CASE WHEN pc.nutriscore_grade = 'd' THEN pc.product_sk END) AS nb_grade_d,
  COUNT(DISTINCT CASE WHEN pc.nutriscore_grade = 'e' THEN pc.product_sk END) AS nb_grade_e,
  COUNT(DISTINCT pc.product_sk) AS nb_produits_total
FROM produits_categories pc
JOIN hive.nutriscan_gold.dim_category dc
  ON pc.category_sk = dc.category_sk
GROUP BY dc.category_name
ORDER BY nb_produits_total DESC
LIMIT 50;
```

Résultats de performance relevés

Indicateur	Valeur
Trino version	435
Queued	3,22 ms
Analysis	415,05 ms
Planning	1,17 s
Execution	35,73 s
Splits traités	63

Données traitées principales

- fact_nutrition scannee : 4 770 925 lignes
- dim_product scannee : 4 521 540 lignes
- dim_product apres filtre nutriscore_grade non NULL : 1 364 541 lignes
- dim_category scan : 35 563 lignes
- Sortie finale : 50 lignes
- Lecture physique fact_nutrition : 20,75 MB
- Lecture physique dim_product : 18,38 MB

Emplacement capture Trino UI - Query Details

 **QUERY DETAILS**

VERSION
435

ENVIRONMENT
PRODUCTION

UPTIME
38.18m

Log Out

20260709_140253_00020_7kpwn

Overview

Live Plan

Stage Performance

Splits

JSON

FINISHED

Session

Useradmin

Principaladmin

SourceApache Superset

Cataloghive

Schema nutriscan_gold

Time zoneUTC

Client Address172.18.0.8

Client Tags

Session Properties

Resource Estimates

Execution

Resource Groupglobal

Submission Time2026-07-09 4:02pm

Completion Time2026-07-09 4:03pm

Elapsed Time36.34s

Queued Time3.22ms

Analysis Time415.05ms

Planning Time1.17s

Execution Time35.92s

Resource Utilization Summary

CPU Time24.52s

Planning CPU Time416.27ms

Scheduled Time2.40m

Input Rows9.33M

Input Data85.2MB

Physical Input Rows9.33M

Physical Input Data40.1MB

Physical Input Read Time964.00ms

Internal Network Rows15.7M

Internal Network Data308MB

Peak User Memory195MB

Peak Total Memory195MB

Cumulative User Memory3.25GB*seconds

Output Rows1.00

Output Data25.0KB

Written Rows0.00

Logical Written Data0B

Physical Written Data0B

Timeline

Parallelism1.05

Scheduled Time/s5.51

Input Rows/s0.00

Input Bytes/s0B

Physical Input Bytes/s0B

Memory Utilization0B

Résultat R4 - EXPLAIN ANALYZE

Trino version: 435 Queued: 3.22ms, Analysis: 415.05ms, Planning: 1.17s, Execution: 35.73s

Fragment 1 [SINGLE]
CPU: 4.17ms, Scheduled: 12.43ms, Blocked 1.70m (Input: 1.14m, Output: 0.00ns), Input: 100 rows (6.97kB); per task: avg.: 100.00 std.dev.: 0.00, Output: 50 rows (3.51kB)
Output layout: [category_name, count_14, count_13, count_12, count_16, count_15, count]
Output partitioning: SINGLE []

```

TopN[count = 50, orderBy = [count_16 DESC NULLS LAST]]
  Layout: [category_name:varchar, count_14:bigint, count_13:bigint, count_12:bigint, count_16:bigint, count_15:bigint, count:bigint]
  Estimates: {rows: 50 (5.32kB), cpu: ?, memory: ?, network: ?}
  CPU: 2.00ms (0.01%), Scheduled: 8.00ms (0.01%), Blocked: 0.00ns (0.00%), Output: 50 rows (3.51kB)
  Input avg.: 100.00 rows, Input std.dev.: 0.00%
└─
LocalExchange[partitioning = SINGLE]
  Layout: [category_name:varchar, count_14:bigint, count_13:bigint, count_12:bigint, count_16:bigint, count_15:bigint, count:bigint]
  Estimates: {rows: ? (?), cpu: 0, memory: 0B, network: 0B}
  CPU: 0.00ns (0.00%), Scheduled: 2.00ms (0.00%), Blocked: 34.09s (3.49%), Output: 100 rows (6.97kB)
  Input avg.: 50.00 rows, Input std.dev.: 0.00%
└─
RemoteSource[sourceFragmentIds = [2]] Layout: [category_name:varchar, count_14:bigint, count_13:bigint, count_12:bigint, count_16:bigint, count_15:bigint, count:bigint]
  CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 1.14m (7.01%), Output: 100 rows (6.97kB)
  Input avg.: 50.00 rows, Input std.dev.: 0.00%
└─
Fragment 2 [HASH]
  CPU: 73.80ms, Scheduled: 685.83ms, Blocked 2.25m (Input: 1.12m, Output: 0.00ns), Input: 23241 rows (1.70MB); per task: avg.: 23241.00 std.dev.: 0.00, Output: 100 rows (6.97kB)
  Output layout: [category_name, count_14, count_13, count_12, count_16, count_15, count]
  Output partitioning: SINGLE []
TopNPartial[count = 50, orderBy = [count_16 DESC NULLS LAST]]
  Layout: [category_name:varchar, count_14:bigint, count_13:bigint, count_12:bigint, count_16:bigint, count_15:bigint, count:bigint]
  CPU: 32.00ms (0.13%), Scheduled: 99.00ms (0.07%), Blocked: 0.00ns (0.00%), Output: 100 rows (6.97kB)
  Input avg.: 10190.00 rows, Input std.dev.: 0.78%
└─
Aggregate[type = FINAL, keys = [category_name]]
  Layout: [category_name:varchar, count_14:bigint, count_13:bigint, count_12:bigint, count_16:bigint, count_15:bigint, count:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
  CPU: 35.00ms (0.14%), Scheduled: 494.00ms (0.34%), Blocked: 0.00ns (0.00%), Output: 20380 rows (1.50MB)
  Input avg.: 11620.50 rows, Input std.dev.: 0.22%
  count_14 := count("count_21")
  count_13 := count("count_22")
  count_12 := count("count_23")
  count_16 := count("count_24")
  count_15 := count("count_25")
  count := count("count_26")
└─
LocalExchange[partitioning = HASH, arguments = ["category_name"]]
  Layout: [category_name:varchar, count_25:bigint, count_24:bigint, count_23:bigint, count_22:bigint, count_26:bigint, count_21:bigint]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 5.00ms (0.02%), Scheduled: 60.00ms (0.04%), Blocked: 1.12m (6.89%), Output: 23241 rows (1.70MB)
  Input avg.: 11620.50 rows, Input std.dev.: 0.11%
└─
RemoteSource[sourceFragmentIds = [3]] Layout: [category_name:varchar, count_25:bigint, count_24:bigint, count_23:bigint, count_22:bigint, count_26:bigint, count_21:bigint]
  CPU: 1.00ms (0.00%), Scheduled: 16.00ms (0.01%), Blocked: 1.12m (6.89%), Output: 23241 rows (1.70MB)
  Input avg.: 11620.50 rows, Input std.dev.: 0.11%
└─
Fragment 3 [HASH]
  CPU: 2.37s, Scheduled: 12.32s, Blocked 1.78m (Input: 54.38s, Output: 0.00ns), Input: 1360980 rows (78.88MB); per task: avg.: 1360980.00 std.dev.: 0.00, Output: 23241 rows (1.70MB)
  Output layout: [category_name, count_25, count_24, count_23, count_22, count_26, count_21]
  Output partitioning: HASH [category_name]
Aggregate[type = PARTIAL, keys = [category_name]]
  Layout: [category_name:varchar, count_25:bigint, count_24:bigint, count_23:bigint, count_22:bigint, count_26:bigint, count_21:bigint]
  CPU: 485.00ms (1.98%), Scheduled: 3.29s (2.29%), Blocked: 0.00ns (0.00%), Output: 23241 rows (1.70MB)
  Input avg.: 680490.00 rows, Input std.dev.: 0.13%
  count_25 := count("expr_11") (mask = expr$distinct_20)
  count_24 := count("product_sk") (mask = product_sk$distinct)
  count_23 := count("expr_8") (mask = expr$distinct_17)
  count_22 := count("expr_9") (mask = expr$distinct_18)
  count_26 := count("expr") (mask = expr$distinct)
  count_21 := count("expr_10") (mask = expr$distinct_19)
└─
MarkDistinct[distinct = [category_name:varchar, product_sk:integer], marker = product_sk$distinct]
  Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:varchar, expr:integer, expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean, expr$distinct_19:boolean, expr$distinct_20:boolean, product_sk$distinct:boolean]
  CPU: 1.56s (6.39%), Scheduled: 6.73s (4.68%), Blocked: 0.00ns (0.00%), Output: 1360980 rows (78.88MB)
  Input avg.: 680490.00 rows, Input std.dev.: 0.13%
└─
LocalExchange[partitioning = HASH, arguments = ["category_name", "product_sk"]]
  Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:varchar, expr:integer, expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean, expr$distinct_19:boolean, expr$distinct_20:boolean]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}

```

```

|
| CPU: 255.00ms (1.04%), Scheduled: 1.98s (1.38%), Blocked: 52.70s (5.40%), Output: 1360980 rows (78.88MB)
|
| Input avg.: 680490.00 rows, Input std.dev.: 19.31%
|
RemoteSource[sourceFragmentIds = [4]] Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer,
category_name:vchar, expr:integer, expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean, expr$distinct_19:boolean,
expr$distinct_20:boolean]
CPU: 68.00ms (0.28%), Scheduled: 306.00ms (0.21%), Blocked: 54.38s (5.57%), Output: 1360980 rows (78.88MB)
Input avg.: 680490.00 rows, Input std.dev.: 19.31%

Fragment 4 [HASH]
CPU: 1.51s, Scheduled: 11.77s, Blocked 1.68m (Input: 51.36s, Output: 0.00ns), Input: 1360980 rows (77.10MB); per task: avg.: 1360980.00 std.dev.:
0.00, Output: 1360980 rows (80.68MB)
Output layout: [expr_11, expr_10, product_sk, expr_9, expr_8, category_name, expr, expr$distinct, expr$distinct_17, expr$distinct_18, expr$distinct_19,
expr$distinct_20]
Output partitioning: HASH [category_name, product_sk]
MarkDistinct[distinct = [category_name:vchar, expr_11:integer], marker = expr$distinct_20]
| Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vchar, expr:integer,
expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean, expr$distinct_19:boolean, expr$distinct_20:boolean]
|
| CPU: 1.10s (4.49%), Scheduled: 8.07s (5.61%), Blocked: 0.00ns (0.00%), Output: 1360980 rows (80.68MB)
|
| Input avg.: 680490.00 rows, Input std.dev.: 24.62%
|
LocalExchange[partitioning = HASH, arguments = ["category_name", "expr_11"]]
| Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vchar, expr:integer,
expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean, expr$distinct_19:boolean]
|
| Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
|
| CPU: 329.00ms (1.35%), Scheduled: 3.09s (2.15%), Blocked: 49.59s (5.08%), Output: 1360980 rows (77.10MB)
|
| Input avg.: 680490.00 rows, Input std.dev.: 18.56%
|
RemoteSource[sourceFragmentIds = [5]] Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer,
category_name:vchar, expr:integer, expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean, expr$distinct_19:boolean]
CPU: 77.00ms (0.31%), Scheduled: 576.00ms (0.40%), Blocked: 51.36s (5.26%), Output: 1360980 rows (77.10MB)
Input avg.: 680490.00 rows, Input std.dev.: 18.56%

Fragment 5 [HASH]
CPU: 1.51s, Scheduled: 11.02s, Blocked 1.57m (Input: 45.70s, Output: 0.00ns), Input: 1360980 rows (74.90MB); per task: avg.: 1360980.00 std.dev.:
0.00, Output: 1360980 rows (78.08MB)
Output layout: [expr_11, expr_10, product_sk, expr_9, expr_8, category_name, expr, expr$distinct, expr$distinct_17, expr$distinct_18, expr$distinct_19]
Output partitioning: HASH [category_name, expr_11]
MarkDistinct[distinct = [category_name:vchar, expr_10:integer], marker = expr$distinct_19]
| Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vchar, expr:integer,
expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean, expr$distinct_19:boolean]
|
| CPU: 1.07s (4.39%), Scheduled: 8.13s (5.65%), Blocked: 0.00ns (0.00%), Output: 1360980 rows (78.08MB)
|
| Input avg.: 680490.00 rows, Input std.dev.: 29.62%
|
LocalExchange[partitioning = HASH, arguments = ["category_name", "expr_10"]]
| Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vchar, expr:integer,
expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean]
|
| Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
|
| CPU: 343.00ms (1.40%), Scheduled: 2.19s (1.52%), Blocked: 48.55s (4.98%), Output: 1360980 rows (74.90MB)
|
| Input avg.: 680490.00 rows, Input std.dev.: 18.57%
|
RemoteSource[sourceFragmentIds = [6]] Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer,
category_name:vchar, expr:integer, expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean]
CPU: 94.00ms (0.38%), Scheduled: 664.00ms (0.46%), Blocked: 45.70s (4.68%), Output: 1360980 rows (74.90MB)
Input avg.: 680490.00 rows, Input std.dev.: 18.57%

Fragment 6 [HASH]
CPU: 1.62s, Scheduled: 11.81s, Blocked 1.64m (Input: 49.15s, Output: 0.00ns), Input: 1360980 rows (72.62MB); per task: avg.: 1360980.00 std.dev.:
0.00, Output: 1360980 rows (75.50MB)
Output layout: [expr_11, expr_10, product_sk, expr_9, expr_8, category_name, expr, expr$distinct, expr$distinct_17, expr$distinct_18]
Output partitioning: HASH [category_name, expr_10]
MarkDistinct[distinct = [category_name:vchar, expr_9:integer], marker = expr$distinct_18]
| Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vchar, expr:integer,
expr$distinct:boolean, expr$distinct_17:boolean, expr$distinct_18:boolean]
|
| CPU: 1.19s (4.85%), Scheduled: 8.82s (6.13%), Blocked: 0.00ns (0.00%), Output: 1360980 rows (75.50MB)
|
| Input avg.: 680490.00 rows, Input std.dev.: 31.25%
|
LocalExchange[partitioning = HASH, arguments = ["category_name", "expr_9"]]
| Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vchar, expr:integer,
expr$distinct:boolean, expr$distinct_17:boolean]
|
| Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
|
| CPU: 340.00ms (1.39%), Scheduled: 2.41s (1.67%), Blocked: 49.54s (5.08%), Output: 1360980 rows (72.62MB)
|
| Input avg.: 680490.00 rows, Input std.dev.: 47.11%
|
RemoteSource[sourceFragmentIds = [7]] Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer,
category_name:vchar, expr:integer, expr$distinct:boolean, expr$distinct_17:boolean]
CPU: 85.00ms (0.35%), Scheduled: 538.00ms (0.37%), Blocked: 49.15s (5.04%), Output: 1360980 rows (72.62MB)
Input avg.: 680490.00 rows, Input std.dev.: 47.11%

```

```

Fragment 7 [HASH]
  CPU: 1.48s, Scheduled: 11.60s, Blocked: 1.55m (Input: 46.62s, Output: 0.00ns), Input: 1360980 rows (70.31MB); per task: avg.: 1360980.00 std.dev.: 0.00, Output: 1360980 rows (72.90MB)
  Output layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr, expr$distinct:boolean, expr$distinct_17:boolean]
  Output partitioning: HASH [category_name:vvarchar, expr:expr]
  MarkDistinct[distinct = [category_name:vvarchar, expr_8:integer], marker = expr$distinct_17]
    Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr, expr$distinct:boolean, expr$distinct_17:boolean]
    CPU: 1.08s (4.43%), Scheduled: 8.82s (6.13%), Blocked: 0.00ns (0.00%), Output: 1360980 rows (72.90MB)
    Input avg.: 680490.00 rows, Input std.dev.: 33.02%
LocalExchange[partitioning = HASH, arguments = ["category_name", "expr_8"]]
  Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr, expr$distinct:boolean]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 309.00ms (1.26%), Scheduled: 2.39s (1.66%), Blocked: 46.70s (4.79%), Output: 1360980 rows (70.31MB)
  Input avg.: 680490.00 rows, Input std.dev.: 52.35%
RemoteSource[sourceFragmentIds = [8]] Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr, expr$distinct:boolean]
  CPU: 82.00ms (0.34%), Scheduled: 384.00ms (0.27%), Blocked: 46.62s (4.78%), Output: 1360980 rows (70.31MB)
  Input avg.: 680490.00 rows, Input std.dev.: 52.35%
Fragment 8 [HASH]
  CPU: 1.82s, Scheduled: 14.94s, Blocked: 1.34m (Input: 41.55s, Output: 0.00ns), Input: 1360980 rows (67.71MB); per task: avg.: 1360980.00 std.dev.: 0.00, Output: 1360980 rows (70.31MB)
  Output layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr, expr$distinct:boolean]
  Output partitioning: HASH [category_name:vvarchar, expr:expr]
  MarkDistinct[distinct = [category_name:vvarchar, expr:expr], marker = expr$distinct]
    Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr, expr$distinct:boolean]
    CPU: 1.28s (5.24%), Scheduled: 10.49s (7.29%), Blocked: 0.00ns (0.00%), Output: 1360980 rows (70.31MB)
    Input avg.: 680490.00 rows, Input std.dev.: 28.39%
LocalExchange[partitioning = HASH, arguments = ["category_name", "expr"]]
  Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 406.00ms (1.66%), Scheduled: 3.55s (2.47%), Blocked: 38.93s (3.99%), Output: 1360980 rows (67.71MB)
  Input avg.: 680490.00 rows, Input std.dev.: 23.90%
RemoteSource[sourceFragmentIds = [9]] Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr]
  CPU: 130.00ms (0.53%), Scheduled: 881.00ms (0.61%), Blocked: 41.55s (4.26%), Output: 1360980 rows (67.71MB)
  Input avg.: 680490.00 rows, Input std.dev.: 23.90%
Fragment 9 [HASH]
  CPU: 3.16s, Scheduled: 18.97s, Blocked: 1.54m (Input: 31.19s, Output: 0.00ns), Input: 1396559 rows (21.69MB); per task: avg.: 1396559.00 std.dev.: 0.00, Output: 1360980 rows (67.71MB)
  Output layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr]
  Output partitioning: HASH [category_name:vvarchar, expr:expr]
Project[]
  Layout: [expr_11:integer, expr_10:integer, product_sk:integer, expr_9:integer, expr_8:integer, category_name:vvarchar, expr:expr]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 2.71s (11.07%), Scheduled: 16.32s (11.35%), Blocked: 0.00ns (0.00%), Output: 1360980 rows (67.71MB)
  Input avg.: 680490.00 rows, Input std.dev.: 5.94%
  expr_11 := (CASE WHEN ("lower" = VARCHAR 'e') THEN "product_sk" END)
  expr_10 := (CASE WHEN ("lower" = VARCHAR 'd') THEN "product_sk" END)
  expr_9 := (CASE WHEN ("lower" = VARCHAR 'c') THEN "product_sk" END)
  expr_8 := (CASE WHEN ("lower" = VARCHAR 'b') THEN "product_sk" END)
  expr := (CASE WHEN ("lower" = VARCHAR 'a') THEN "product_sk" END)
InnerJoin[criteria = ("category_sk" = "category_sk_4"), distribution = PARTITIONED]
  Layout: [product_sk:integer, lower:vvarchar, category_name:vvarchar]
  Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
  CPU: 435.00ms (1.78%), Scheduled: 2.54s (1.77%), Blocked: 1.02m (6.27%), Output: 1360980 rows (43.99MB)
  Left (probe)
    Input avg.: 680498.00 rows, Input std.dev.: 5.94%
  Right (build)
    Input avg.: 17781.50 rows, Input std.dev.: 0.37%
    Distribution: PARTITIONED
  dynamicFilterAssignments = {category_sk_4 -> #df_1110}
RemoteSource[sourceFragmentIds = [10]]
  Layout: [product_sk:integer, category_sk:integer, lower:vvarchar]
  CPU: 17.00ms (0.07%), Scheduled: 45.00ms (0.03%), Blocked: 30.95s (3.17%), Output: 1360996 rows (20.77MB)
  Input avg.: 680498.00 rows, Input std.dev.: 5.94%
LocalExchange[partitioning = HASH, arguments = ["category_sk_4"]]

```

```

    | Layout: [category_sk_4:integer, category_name:varchar]
    | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    | CPU: 3.00ms (0.01%), Scheduled: 57.00ms (0.04%), Blocked: 525.00ms (0.05%), Output: 35563 rows (943.60kB)
    | Input avg.: 17781.50 rows, Input std.dev.: 100.00%
    |
RemoteSource[sourceFragmentIds = [13]] Layout: [category_sk_4:integer, category_name:varchar]
  CPU: 1.00ms (0.00%), Scheduled: 1.00ms (0.00%), Blocked: 243.00ms (0.02%), Output: 35563 rows (943.60kB)
  Input avg.: 17781.50 rows, Input std.dev.: 100.00%

Fragment 10 [HASH]
  CPU: 8.96s, Scheduled: 40.87s, Blocked 1.18m (Input: 9.95s, Output: 0.00ns), Input: 6135466 rows (59.81MB); per task: avg.: 6135466.00 std.dev.: 0.00,
  Output: 1360996 rows (20.77MB)
  Output layout: [product_sk, category_sk, lower]
  Output partitioning: HASH [category_sk]
  Aggregate[type = FINAL, keys = [product_sk, category_sk, lower]]
    | Layout: [product_sk:integer, category_sk:integer, lower:varchar]
    | Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
    | CPU: 2.95s (12.08%), Scheduled: 18.34s (12.75%), Blocked: 0.00ns (0.00%), Output: 1360996 rows (20.77MB)
    | Input avg.: 691555.00 rows, Input std.dev.: 0.20%
    |
LocalExchange[partitioning = HASH, arguments = ["product_sk", "category_sk", "lower"]]
  | Layout: [product_sk:integer, category_sk:integer, lower:varchar]
  | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  | CPU: 311.00ms (1.27%), Scheduled: 1.16s (0.80%), Blocked: 22.61s (2.32%), Output: 1383110 rows (21.10MB)
  | Input avg.: 691555.00 rows, Input std.dev.: 14.83%
  |
Aggregate[type = PARTIAL, keys = [product_sk, category_sk, lower]]
  | Layout: [product_sk:integer, category_sk:integer, lower:varchar]
  | CPU: 3.03s (12.40%), Scheduled: 11.96s (8.31%), Blocked: 0.00ns (0.00%), Output: 1383110 rows (22.17MB)
  | Input avg.: 769366.00 rows, Input std.dev.: 15.83%
  |
Project[]
  | Layout: [category_sk:integer, product_sk:integer, lower:varchar]
  | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  | CPU: 392.00ms (1.60%), Scheduled: 1.20s (0.83%), Blocked: 0.00ns (0.00%), Output: 1538732 rows (35.21MB)
  | Input avg.: 769366.00 rows, Input std.dev.: 15.83%
  | lower := lower("nutriscore_grade")
  |
InnerJoin[criteria = ("product_sk" = "product_sk_0"), distribution = PARTITIONED]
  | Layout: [product_sk:integer, category_sk:integer, nutriscore_grade:varchar]
  | Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
  | CPU: 1.96s (8.01%), Scheduled: 6.64s (4.62%), Blocked: 29.20s (2.99%), Output: 1538732 rows (35.22MB)
  | Left (probe)
  | Input avg.: 2385462.50 rows, Input std.dev.: 9.89%
  | Right (build)
  | Input avg.: 682270.50 rows, Input std.dev.: 0.05%
  | Distribution: PARTITIONED
  | maySkipOutputDuplicates = true
  | dynamicFilterAssignments = {product_sk_0 -> #df_1111}
  |
RemoteSource[sourceFragmentIds = [11]]
  | Layout: [product_sk:integer, category_sk:integer]
  | CPU: 96.00ms (0.39%), Scheduled: 198.00ms (0.14%), Blocked: 1.41s (0.14%), Output: 4770925 rows (45.50MB)
  | Input avg.: 2385462.50 rows, Input std.dev.: 9.89%
  |
LocalExchange[partitioning = HASH, arguments = ["product_sk_0"]]
  | Layout: [product_sk_0:integer, nutriscore_grade:varchar]
  | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  | CPU: 145.00ms (0.59%), Scheduled: 977.00ms (0.68%), Blocked: 9.17s (0.94%), Output: 1364541 rows (14.31MB)
  | Input avg.: 682270.50 rows, Input std.dev.: 0.26%
  |
RemoteSource[sourceFragmentIds = [12]] Layout: [product_sk_0:integer, nutriscore_grade:varchar]
  CPU: 64.00ms (0.26%), Scheduled: 332.00ms (0.23%), Blocked: 8.54s (0.88%), Output: 1364541 rows (14.31MB)
  Input avg.: 682270.50 rows, Input std.dev.: 0.26%

Fragment 11 [SOURCE]
  CPU: 954.81ms, Scheduled: 4.90s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 4770925 rows (45.50MB); per task: avg.: 4770925.00 std.dev.:
  0.00, Output: 4770925 rows (45.50MB)
  Output layout: [product_sk, category_sk]
  Output partitioning: HASH [product_sk]
  ScanFilter[table = hive:nutriscan_gold:fact_nutrition, dynamicFilters = {"category_sk" = #df_1110, "product_sk" = #df_1111}] Layout: [product_sk:integer,
  category_sk:integer]
  | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  | CPU: 953.00ms (3.90%), Scheduled: 4.89s (3.40%), Blocked: 0.00ns (0.00%), Output: 4770925 rows (45.50MB)

```

Input avg.: 681560.71 rows, Input std.dev.: 244.95% category_sk := category_sk:int:REGULAR product_sk := product_sk:int:REGULAR Input: 4770925 rows (45.50MB), Filtered: 0.00%, Physical input: 20.75MB, Physical input time: 449.05ms

Dynamic filters:

- df_1110, ALL, collection time=796.53ms
- df_1111, ALL, collection time=5.21s

Fragment 12 [SOURCE]

CPU: 999.53ms, Scheduled: 4.83s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 4521540 rows (38.83MB); per task: avg.: 4521540.00 std.dev.: 0.00, Output: 1364541 rows (11.78MB)

Output layout: [product_sk_0, nutriscore_grade]

Output partitioning: HASH [product_sk_0]

ScanFilter[table = hive:nutriscan_gold:dim_product, filterPredicate = (NOT ("nutriscore_grade" IS NULL))] Layout: [product_sk_0:integer, nutriscore_grade:varchar]

Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}

CPU: 998.00ms (4.08%), Scheduled: 4.83s (3.36%), Blocked: 0.00ns (0.00%), Output: 1364541 rows (11.78MB)

Input avg.: 645934.29 rows, Input std.dev.: 244.95% product_sk_0 := product_sk:int:REGULAR nutriscore_grade := nutriscore_grade:string:REGULAR Input: 4521540 rows (38.83MB), Filtered: 69.82%, Physical input: 18.38MB, Physical input time: 432.56ms

Fragment 13 [SOURCE]

CPU: 25.15ms, Scheduled: 367.98ms, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 35563 rows (943.60kB); per task: avg.: 35563.00 std.dev.: 0.00, Output: 35563 rows (943.60kB)

Output layout: [category_sk_4, category_name]

Output partitioning: HASH [category_sk_4]

TableScan[table = hive:nutriscan_gold:dim_category] Layout: [category_sk_4:integer, category_name:varchar]

Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}

CPU: 24.00ms (0.10%), Scheduled: 367.00ms (0.26%), Blocked: 0.00ns (0.00%), Output: 35563 rows (943.60kB)

Input avg.: 35563.00 rows, Input std.dev.: 0.00% category_name := category_name:string:REGULAR category_sk_4 := category_sk:int:REGULAR Input: 35563 rows (943.60kB), Physical input: 964.47kB, Physical input time: 82.31ms

Requete 5 - SLICE France, top marques par sel moyen

Objectif : Filtrer les produits vendus en France puis identifier les 10 marques avec le sel moyen le plus eleve.

Type OLAP : SLICE / filtrage dimensionnel

Requête exécutée

```
EXPLAIN ANALYZE
WITH produits_france AS (
  SELECT DISTINCT
    fn.product_sk,
    fn.brand_sk,
    fn.salt_g
  FROM hive.nutriscan_gold.fact_nutrition fn
  JOIN hive.nutriscan_gold.dim_country dco
    ON fn.country_sk = dco.country_sk
  WHERE dco.country_tag = 'en:france'
    AND fn.salt_g IS NOT NULL
    AND fn.brand_sk IS NOT NULL
)
SELECT
  db.brand_name,
  ROUND(AVG(pf.salt_g), 3) AS sel_moyen_g,
  COUNT(DISTINCT pf.product_sk) AS nb_produits
FROM produits_france pf
JOIN hive.nutriscan_gold.dim_brand db
  ON pf.brand_sk = db.brand_sk
GROUP BY db.brand_name
ORDER BY sel_moyen_g DESC
LIMIT 10;
```

Résultats de performance relevés

Indicateur	Valeur
Trino version	435
Queued	9,16 ms
Analysis	427,20 ms
Planning	1,07 s
Execution	11,86 s
Splits traités	39

Données traitées principales

- fact_nutrition scannee : 4 770 925 lignes
- Filtre country_tag = en:france applique via dim_country
- Sortie finale : 10 lignes
- Lecture physique fact_nutrition : 33,11 MB
- fact_nutrition filtree : 53,61 % des lignes filtrees

Emplacement capture Trino UI - Query Details

 **QUERY DETAILS**

VERSION

435

ENVIRONMENT

PRODUCTION

UPTIME

39.49m

Log Out

20260709_140437_00021_7kpwn

Overview

Live Plan

Stage Performance

Splits

JSON

FINISHED

Session

User	admin
Principal	admin
Source	Apache Superset
Catalog	hive
Schema	nutriscan_gold
Time zone	UTC
Client Address	172.18.0.8
Client Tags	
Session Properties	
Resource Estimates	

Execution

Resource Group	global
Submission Time	2026-07-09 4:04pm
Completion Time	2026-07-09 4:04pm
Elapsed Time	12.42s
Queued Time	9.16ms
Analysis Time	427.20ms
Planning Time	1.07s
Execution Time	11.98s

Resource Utilization Summary

CPU Time	6.87s
Planning CPU Time	367.38ms
Scheduled Time	27.05s
Input Rows	5.12M
Input Data	98.7MB
Physical Input Rows	5.12M
Physical Input Data	37.6MB
Physical Input Read Time	651.00ms
Internal Network Rows	4.17M
Internal Network Data	66.0MB
Peak User Memory	63.0MB
Peak Total Memory	63.0MB
Cumulative User Memory	369MB*seconds
Output Rows	1.00
Output Data	13.9KB
Written Rows	0.00
Logical Written Data	0B
Physical Written Data	0B

Timeline

Parallelism	0.70
Scheduled Time/s	2.86
Input Rows/s	0.00
Input Bytes/s	0B
Physical Input Bytes/s	0B
Memory Utilization	0B

Résultat R5- EXPLAIN ANALYZE

```
Trino version: 435 Queued: 9.16ms, Analysis: 427.20ms, Planning: 1.07s, Execution: 11.86s
Fragment 1 [SINGLE]
  CPU: 5.17ms, Scheduled: 9.58ms, Blocked 30.79s (Input: 20.54s, Output: 0.00ns), Input: 20 rows (617B); per task: avg.: 20.00 std.dev.: 0.00, Output: 10 rows (296B)
Output layout: [round, brand_name, count]
Output partitioning: SINGLE []
TopN[count = 10, orderBy = [round DESC NULLS LAST]]
```

```

    | Layout: [round:real, brand_name:varchar, count:bigint]
    | Estimates: {rows: 10 (690B), cpu: ?, memory: ?, network: ?}
    | CPU: 1.00ms (0.01%), Scheduled: 3.00ms (0.01%), Blocked: 0.00ns (0.00%), Output: 10 rows (296B)
    | Input avg.: 20.00 rows, Input std.dev.: 0.00%
    |
LocalExchange[partitioning = SINGLE]
    | Layout: [round:real, brand_name:varchar, count:bigint]
    | Estimates: {rows: ? (?), cpu: 0, memory: 0B, network: 0B}
    | CPU: 1.00ms (0.01%), Scheduled: 4.00ms (0.01%), Blocked: 10.27s (6.65%), Output: 20 rows (617B)
    | Input avg.: 10.00 rows, Input std.dev.: 0.00%
    |
RemoteSource[sourceFragmentIds = [2]] Layout: [round:real, brand_name:varchar, count:bigint]
    | CPU: 1.00ms (0.01%), Scheduled: 1.00ms (0.00%), Blocked: 20.54s (13.30%), Output: 20 rows (617B)
    | Input avg.: 10.00 rows, Input std.dev.: 0.00%
    |
Fragment 2 [HASH]
    | CPU: 469.26ms, Scheduled: 2.07s, Blocked 34.79s (Input: 17.53s, Output: 0.00ns), Input: 72825 rows (3.13MB); per task: avg.: 72825.00 std.dev.: 0.00,
    | Output: 20 rows (617B)
    | Output layout: [round, brand_name, count]
    | Output partitioning: SINGLE []
    | TopNPartial[count = 10, orderBy = [round DESC NULLS LAST]]
    | | Layout: [round:real, brand_name:varchar, count:bigint]
    | | CPU: 72.00ms (1.06%), Scheduled: 264.00ms (0.98%), Blocked: 0.00ns (0.00%), Output: 20 rows (617B)
    | | Input avg.: 27452.50 rows, Input std.dev.: 0.24%
    | |
    | Project[]
    | | Layout: [round:real, brand_name:varchar, count:bigint]
    | | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    | | CPU: 81.00ms (1.19%), Scheduled: 311.00ms (1.16%), Blocked: 0.00ns (0.00%), Output: 54905 rows (1.65MB)
    | | Input avg.: 27452.50 rows, Input std.dev.: 0.24%
    | | round := round("avg", 3)
    | |
    | Aggregate[type = FINAL, keys = [brand_name]]
    | | Layout: [brand_name:varchar, avg:real, count:bigint]
    | | Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
    | | CPU: 245.00ms (3.59%), Scheduled: 1.12s (4.16%), Blocked: 0.00ns (0.00%), Output: 54905 rows (1.65MB)
    | | Input avg.: 36412.50 rows, Input std.dev.: 0.26%
    | | avg := avg("avg_8")
    | | count := count("count_9")
    | |
    | LocalExchange[partitioning = HASH, arguments = ["brand_name"]]
    | | Layout: [brand_name:varchar, avg_8:row(bigint, double), count_9:bigint]
    | | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    | | CPU: 52.00ms (0.76%), Scheduled: 295.00ms (1.10%), Blocked: 17.28s (11.19%), Output: 72825 rows (3.13MB)
    | | Input avg.: 36412.50 rows, Input std.dev.: 58.20%
    | |
    | RemoteSource[sourceFragmentIds = [3]] Layout: [brand_name:varchar, avg_8:row(bigint, double), count_9:bigint]
    | | CPU: 18.00ms (0.26%), Scheduled: 80.00ms (0.30%), Blocked: 17.53s (11.35%), Output: 72825 rows (3.13MB)
    | | Input avg.: 36412.50 rows, Input std.dev.: 58.20%
    | |
    | Fragment 3 [HASH]
    | | CPU: 1.34s, Scheduled: 4.85s, Blocked 28.23s (Input: 14.82s, Output: 0.00ns), Input: 513408 rows (12.12MB); per task: avg.: 513408.00 std.dev.: 0.00,
    | | Output: 72825 rows (3.41MB)
    | | Output layout: [brand_name, avg_8, count_9]
    | | Output partitioning: HASH [brand_name]
    | | Aggregate[type = PARTIAL, keys = [brand_name]]
    | | | Layout: [brand_name:varchar, avg_8:row(bigint, double), count_9:bigint]
    | | | CPU: 646.00ms (9.47%), Scheduled: 2.38s (8.86%), Blocked: 0.00ns (0.00%), Output: 72825 rows (3.41MB)
    | | | Input avg.: 256704.00 rows, Input std.dev.: 0.03%
    | | | avg_8 := avg("salt_g")
    | | | count_9 := count("product_sk") (mask = product_sk$distinct)
    | | |
    | | MarkDistinct[distinct = [brand_name:varchar, product_sk:integer], marker = product_sk$distinct]
    | | | Layout: [product_sk:integer, salt_g:real, brand_name:varchar, product_sk$distinct:boolean]
    | | | CPU: 550.00ms (8.06%), Scheduled: 1.72s (6.43%), Blocked: 0.00ns (0.00%), Output: 513408 rows (12.12MB)
    | | | Input avg.: 256704.00 rows, Input std.dev.: 0.03%
    | | |
    | | LocalExchange[partitioning = HASH, arguments = ["brand_name", "product_sk"]]
    | | | Layout: [product_sk:integer, salt_g:real, brand_name:varchar]
    | | | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    | | | CPU: 127.00ms (1.86%), Scheduled: 690.00ms (2.57%), Blocked: 13.44s (8.70%), Output: 513408 rows (12.12MB)
    | | | Input avg.: 256704.00 rows, Input std.dev.: 23.25%
    | | |

```

```

└─
RemoteSource[sourceFragmentIds = [4]] Layout: [product_sk:integer, salt_g:real, brand_name:varchar]
CPU: 15.00ms (0.22%), Scheduled: 64.00ms (0.24%), Blocked: 14.82s (9.59%), Output: 513408 rows (12.12MB)
Input avg.: 256704.00 rows, Input std.dev.: 23.25%

Fragment 4 [HASH]
CPU: 594.76ms, Scheduled: 2.86s, Blocked 31.08s (Input: 14.83s, Output: 0.00ns), Input: 862183 rows (15.03MB); per task: avg.: 862183.00 std.dev.: 0.00, Output: 513408 rows (12.12MB)
Output layout: [product_sk, salt_g, brand_name]
Output partitioning: HASH [brand_name, product_sk]
InnerJoin[criteria = ("brand_sk" = "brand_sk_4"), distribution = PARTITIONED]
└─ Layout: [product_sk:integer, salt_g:real, brand_name:varchar]
└─ Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
└─ CPU: 536.00ms (7.85%), Scheduled: 2.61s (9.75%), Blocked: 14.75s (9.55%), Output: 513408 rows (12.12MB)
└─ Left (probe)
└─ Input avg.: 256704.00 rows, Input std.dev.: 0.06%
└─ Right (build)
└─ Input avg.: 174387.50 rows, Input std.dev.: 0.21%
└─ Distribution: PARTITIONED
└─ dynamicFilterAssignments = {brand_sk_4 -> #df_1059}
└─
RemoteSource[sourceFragmentIds = [5]]
└─ Layout: [product_sk:integer, brand_sk:integer, salt_g:real]
└─ CPU: 11.00ms (0.16%), Scheduled: 59.00ms (0.22%), Blocked: 13.46s (8.71%), Output: 513408 rows (7.34MB)
└─ Input avg.: 256704.00 rows, Input std.dev.: 0.06%
└─
LocalExchange[partitioning = HASH, arguments = ["brand_sk_4"]]
└─ Layout: [brand_sk_4:integer, brand_name:varchar]
└─ Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
└─ CPU: 39.00ms (0.57%), Scheduled: 164.00ms (0.61%), Blocked: 1.51s (0.98%), Output: 348775 rows (7.69MB)
└─ Input avg.: 174387.50 rows, Input std.dev.: 24.84%
└─
RemoteSource[sourceFragmentIds = [9]] Layout: [brand_sk_4:integer, brand_name:varchar]
CPU: 6.00ms (0.09%), Scheduled: 20.00ms (0.07%), Blocked: 1.37s (0.89%), Output: 348775 rows (7.69MB)
Input avg.: 174387.50 rows, Input std.dev.: 24.84%

Fragment 5 [HASH]
CPU: 1.09s, Scheduled: 4.39s, Blocked 23.52s (Input: 12.83s, Output: 0.00ns), Input: 513408 rows (7.34MB); per task: avg.: 513408.00 std.dev.: 0.00, Output: 513408 rows (7.34MB)
Output layout: [product_sk, brand_sk, salt_g]
Output partitioning: HASH [brand_sk]
Aggregate[type = FINAL, keys = [product_sk, brand_sk, salt_g]]
└─ Layout: [product_sk:integer, brand_sk:integer, salt_g:real]
└─ Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
└─ CPU: 990.00ms (14.51%), Scheduled: 3.81s (14.22%), Blocked: 0.00ns (0.00%), Output: 513408 rows (7.34MB)
└─ Input avg.: 256704.00 rows, Input std.dev.: 0.06%
└─
LocalExchange[partitioning = HASH, arguments = ["product_sk", "brand_sk", "salt_g"]]
└─ Layout: [product_sk:integer, brand_sk:integer, salt_g:real]
└─ Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
└─ CPU: 85.00ms (1.25%), Scheduled: 523.00ms (1.95%), Blocked: 10.69s (6.92%), Output: 513408 rows (7.34MB)
└─ Input avg.: 256704.00 rows, Input std.dev.: 4.69%
└─
RemoteSource[sourceFragmentIds = [6]] Layout: [product_sk:integer, brand_sk:integer, salt_g:real]
CPU: 9.00ms (0.13%), Scheduled: 30.00ms (0.11%), Blocked: 12.83s (8.31%), Output: 513408 rows (7.34MB)
Input avg.: 256704.00 rows, Input std.dev.: 4.69%

Fragment 6 [HASH]
CPU: 1.64s, Scheduled: 7.22s, Blocked 14.78s (Input: 5.23s, Output: 0.00ns), Input: 2212996 rows (42.21MB); per task: avg.: 2212996.00 std.dev.: 0.00, Output: 513408 rows (7.34MB)
Output layout: [product_sk, brand_sk, salt_g]
Output partitioning: HASH [product_sk, brand_sk, salt_g]
Aggregate[type = PARTIAL, keys = [product_sk, brand_sk, salt_g]]
└─ Layout: [product_sk:integer, brand_sk:integer, salt_g:real]
└─ CPU: 1.20s (17.53%), Scheduled: 4.93s (18.39%), Blocked: 0.00ns (0.00%), Output: 513408 rows (7.34MB)
└─ Input avg.: 256704.00 rows, Input std.dev.: 32.82%
└─
InnerJoin[criteria = ("country_sk" = "country_sk_0"), distribution = PARTITIONED]
└─ Layout: [product_sk:integer, brand_sk:integer, salt_g:real]
└─ Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
└─ CPU: 397.00ms (5.82%), Scheduled: 1.76s (6.55%), Blocked: 0.00ns (0.00%), Output: 513408 rows (13.22MB)
└─ Left (probe)
└─ Input avg.: 1106497.50 rows, Input std.dev.: 26.19%
└─ Right (build)
└─ Input avg.: 0.50 rows, Input std.dev.: 100.00%
└─ Distribution: PARTITIONED
└─ maySkipOutputDuplicates = true

```

```

dynamicFilterAssignments = {country_sk_0 -> #df_1060}
├─
RemoteSource[sourceFragmentIds = [7]]
  Layout: [product_sk:integer, brand_sk:integer, country_sk:integer, salt_g:real]
  CPU: 38.00ms (0.56%), Scheduled: 502.00ms (1.87%), Blocked: 4.70s (3.04%), Output: 2212995 rows (42.21MB)
  Input avg.: 1106497.50 rows, Input std.dev.: 26.19%
├─
LocalExchange[partitioning = HASH, arguments = ["country_sk_0"]]
  Layout: [country_sk_0:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 0.00ms (0.00%), Scheduled: 2.00ms (0.01%), Blocked: 762.00ms (0.49%), Output: 1 row (5B)
  Input avg.: 0.50 rows, Input std.dev.: 100.00%
├─
RemoteSource[sourceFragmentIds = [8]] Layout: [country_sk_0:integer]
  CPU: 0.00ms (0.00%), Scheduled: 0.00ms (0.00%), Blocked: 531.00ms (0.34%), Output: 1 row (5B)
  Input avg.: 0.50 rows, Input std.dev.: 100.00%
├─
Fragment 7 [SOURCE]
  CPU: 1.49s, Scheduled: 4.92s, Blocked 0.00ms (Input: 0.00ms, Output: 0.00ms), Input: 4770925 rows (91.00MB); per task: avg.: 4770925.00 std.dev.: 0.00, Output: 2212995 rows (42.21MB)
  Output layout: [product_sk, brand_sk, country_sk, salt_g]
  Output partitioning: HASH [country_sk]
  ScanFilter[table = hive:nutriscan_gold:fact_nutrition, filterPredicate = ((NOT ("salt_g" IS NULL)) AND (NOT ("brand_sk" IS NULL))), dynamicFilters = {"brand_sk" = #df_1059, "country_sk" = #df_1060}] Layout: [product_sk:integer, brand_sk:integer, country_sk:integer, salt_g:real]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 1.49s (21.89%), Scheduled: 4.90s (18.28%), Blocked: 0.00ms (0.00%), Output: 2212995 rows (42.21MB)
  Input avg.: 681560.71 rows, Input std.dev.: 244.95% product_sk := product_sk:int:REGULAR salt_g := salt_g:float:REGULAR country_sk := country_sk:int:REGULAR brand_sk := brand_sk:int:REGULAR Input: 4770925 rows (91.00MB), Filtered: 53.61%, Physical input: 33.11MB, Physical input time: 503.09ms
  Dynamic filters:
    - df_1059, ALL, collection time=1.19s
    - df_1060, [ SortedRangeSet[type=integer, ranges=1, {[118]}] ], collection time=938.51ms
├─
Fragment 8 [SOURCE]
  CPU: 19.44ms, Scheduled: 69.55ms, Blocked 0.00ms (Input: 0.00ms, Output: 0.00ms), Input: 384 rows (7.19kB); per task: avg.: 384.00 std.dev.: 0.00, Output: 1 row (5B)
  Output layout: [country_sk_0]
  Output partitioning: HASH [country_sk_0]
  ScanFilterProject[table = hive:nutriscan_gold:dim_country, filterPredicate = ("country_tag" = VARCHAR 'en:france')] Layout: [country_sk_0:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 19.00ms (0.28%), Scheduled: 69.00ms (0.26%), Blocked: 0.00ms (0.00%), Output: 1 row (5B)
  Input avg.: 384.00 rows, Input std.dev.: 0.00% country_tag := country_tag:string:REGULAR country_sk_0 := country_sk:int:REGULAR Input: 384 rows (7.19kB), Filtered: 99.74%, Physical input: 10.63kB, Physical input time: 32.82ms
├─
Fragment 9 [SOURCE]
  CPU: 194.72ms, Scheduled: 503.30ms, Blocked 0.00ms (Input: 0.00ms, Output: 0.00ms), Input: 348775 rows (7.69MB); per task: avg.: 348775.00 std.dev.: 0.00, Output: 348775 rows (7.69MB)
  Output layout: [brand_sk_4, brand_name]
  Output partitioning: HASH [brand_sk_4]
  ScanFilter[table = hive:nutriscan_gold:dim_brand, filterPredicate = (NOT ("brand_sk_4" IS NULL))] Layout: [brand_sk_4:integer, brand_name:varchar]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 195.00ms (2.86%), Scheduled: 503.00ms (1.88%), Blocked: 0.00ms (0.00%), Output: 348775 rows (7.69MB)
  Input avg.: 348775.00 rows, Input std.dev.: 0.00% brand_name := brand_name:string:REGULAR brand_sk_4 := brand_sk:int:REGULAR Input: 348775 rows (7.69MB), Filtered: 0.00%, Physical input: 4.48MB, Physical input time: 115.05ms

```

Requete 6 - Fenetre glissante 3 mois

Objectif : Calculer le Nutri-Score moyen mensuel, puis une moyenne mobile sur trois mois glissants.

Type OLAP : Fenetre glissante / moyenne mobile temporelle

Requête exécutée

```
EXPLAIN ANALYZE
WITH mensuel AS (
  SELECT
    dd.year,
    dd.month,
    CONCAT(CAST(dd.year AS VARCHAR), '-', LPAD(CAST(dd.month AS VARCHAR), 2, '0')) AS periode,
    ROUND(AVG(fn.nutriscore_score), 2) AS nutriscore_moyen_mois,
    COUNT(DISTINCT fn.product_sk) AS nb_produits
  FROM hive.nutriscan_gold.fact_nutrition fn
  JOIN hive.nutriscan_gold.dim_date dd
    ON fn.date_sk = dd.date_sk
  WHERE fn.nutriscore_score IS NOT NULL
  GROUP BY dd.year, dd.month
)
SELECT
  periode,
  year,
  month,
  nutriscore_moyen_mois,
  nb_produits,
  ROUND(
    AVG(nutriscore_moyen_mois) OVER (
      ORDER BY year, month
      ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
    ),
    2
  ) AS moyenne_mobile_3_mois
FROM mensuel
ORDER BY year, month;
```

Résultats de performance relevés

Indicateur	Valeur
Trino version	435
Queued	5,58 ms
Analysis	280,08 ms
Planning	1,61 s
Execution	6,57 s
Splits traités	31

Données traitées principales

- fact_nutrition scannée : 4 770 925 lignes
- Lignes avec nutriscore_score renseigné : 1 538 741
- dim_date scan : 3 856 lignes
- Sortie finale : 127 lignes
- Lecture physique fact_nutrition : 26,17 MB

 QUERY DETAILS

VERSION
435

ENVIRONMENT
PRODUCTION

UPTIME
40.15m

Log Out

20260709_140511_00022_7kpwn

Overview

Live Plan

Stage Performance

Splits

JSON

FINISHED

Session

User	admin
Principal	admin
Source	Apache Superset
Catalog	hive
Schema	nutriscan_gold
Time zone	UTC
Client Address	172.18.0.8
Client Tags	
Session Properties	
Resource Estimates	

Execution

Resource Group	global
Submission Time	2026-07-09 4:05pm
Completion Time	2026-07-09 4:05pm
Elapsed Time	6.91s
Queued Time	5.58ms
Analysis Time	280.08ms
Planning Time	1.61s
Execution Time	6.63s

Resource Utilization Summary

CPU Time	2.94s
Planning CPU Time	353.51ms
Scheduled Time	15.07s
Input Rows	4.77M
Input Data	68.3MB
Physical Input Rows	4.77M
Physical Input Data	26.2MB
Physical Input Read Time	397.00ms
Internal Network Rows	3.08M
Internal Network Data	47.0MB
Peak User Memory	63.2MB
Peak Total Memory	63.2MB
Cumulative User Memory	169MB*seconds
Output Rows	1.00
Output Data	12.0KB
Written Rows	0.00
Logical Written Data	0B
Physical Written Data	0B

Timeline

Parallelism	0.43
Scheduled Time/s	2.18
Input Rows/s	691K
Input Bytes/s	9.88MB
Physical Input Bytes/s	66.0MB
Memory Utilization	0B

Résultat R6 - EXPLAIN ANALYZE

```
T rino version: 435 Queued: 5.58ms, Analysis: 280.08ms, Planning: 1.61s, Execution: 6.57s

Fragment 1 [ROUND_ROBIN]
  CPU: 15.33ms, Scheduled: 49.80ms, Blocked 13.59s (Input: 9.06s, Output: 0.00ns), Input: 127 rows (6.08kB); per task: avg.: 127.00 std.dev.: 0.00,
  Output: 127 rows (6.08kB)
  Output layout: [round_7, round, count, year, concat, month]
  Output partitioning: SINGLE []
  LocalMerge[orderBy = [year ASC NULLS LAST, month ASC NULLS LAST]]
```

```

    | Layout: [round_7:double, round:double, count:bigint, year:integer, concat:vchar, month:integer]
    | Estimates: {rows: ? (?), cpu: 0, memory: 0B, network: 0B}
    | CPU: 4.00ms (0.14%), Scheduled: 12.00ms (0.08%), Blocked: 4.54s (6.86%), Output: 127 rows (6.08kB)
    | Input avg.: 63.50 rows, Input std.dev.: 100.00%
    |
PartialSort[orderBy = [year ASC NULLS LAST, month ASC NULLS LAST]]
    | Layout: [round_7:double, round:double, count:bigint, year:integer, concat:vchar, month:integer]
    | CPU: 10.00ms (0.35%), Scheduled: 32.00ms (0.22%), Blocked: 0.00ns (0.00%), Output: 127 rows (6.08kB)
    | Input avg.: 63.50 rows, Input std.dev.: 100.00%
    |
RemoteSource[sourceFragmentIds = [2]] Layout: [round_7:double, round:double, count:bigint, year:integer, concat:vchar, month:integer]
    CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 9.06s (13.68%), Output: 127 rows (6.08kB)
    Input avg.: 63.50 rows, Input std.dev.: 100.00%

Fragment 2 [SINGLE]
    CPU: 19.89ms, Scheduled: 50.54ms, Blocked 21.99s (Input: 8.78s, Output: 0.00ns), Input: 127 rows (6.70kB); per task: avg.: 127.00 std.dev.: 0.00,
    Output: 127 rows (6.08kB)
    Output layout: [round_7, round, count, year, concat, month]
    Output partitioning: ROUND_ROBIN []
Project[]
    | Layout: [round_7:double, round:double, count:bigint, year:integer, concat:vchar, month:integer]
    | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    | CPU: 3.00ms (0.10%), Scheduled: 20.00ms (0.13%), Blocked: 0.00ns (0.00%), Output: 127 rows (6.08kB)
    | Input avg.: 63.50 rows, Input std.dev.: 100.00%
    | round_7 := round("avg_6", "expr_4")
    |
LocalExchange[partitioning = ROUND_ROBIN]
    | Layout: [expr_5:bigint, expr_4:integer, month:integer, round:double, year:integer, count:bigint, concat:vchar, avg_6:double]
    | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    | CPU: 0.00ns (0.00%), Scheduled: 2.00ms (0.01%), Blocked: 8.87s (13.39%), Output: 127 rows (7.81kB)
    | Input avg.: 127.00 rows, Input std.dev.: 0.00%
    |
Window[orderBy = [year ASC NULLS LAST, month ASC NULLS LAST]]
    | Layout: [expr_5:bigint, expr_4:integer, month:integer, round:double, year:integer, count:bigint, concat:vchar, avg_6:double]
    | CPU: 13.00ms (0.45%), Scheduled: 26.00ms (0.18%), Blocked: 0.00ns (0.00%), Output: 127 rows (7.81kB)
    | Input avg.: 127.00 rows, Input std.dev.: 0.00%
    | avg_6 := avg("round") ROWS expr_5 PRECEDING CURRENT_ROW
    |
LocalExchange[partitioning = SINGLE]
    | Layout: [expr_5:bigint, expr_4:integer, month:integer, round:double, year:integer, count:bigint, concat:vchar]
    | Estimates: {rows: ? (?), cpu: 0, memory: 0B, network: 0B}
    | CPU: 0.00ns (0.00%), Scheduled: 1.00ms (0.01%), Blocked: 4.32s (6.52%), Output: 127 rows (6.70kB)
    | Input avg.: 63.50 rows, Input std.dev.: 13.39%
    |
RemoteSource[sourceFragmentIds = [3]] Layout: [expr_5:bigint, expr_4:integer, month:integer, round:double, year:integer, count:bigint, concat:vchar]
    CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 8.78s (13.26%), Output: 127 rows (6.70kB)
    Input avg.: 63.50 rows, Input std.dev.: 13.39%

Fragment 3 [HASH]
    CPU: 15.05ms, Scheduled: 47.14ms, Blocked 17.07s (Input: 8.54s, Output: 0.00ns), Input: 238 rows (8.83kB); per task: avg.: 238.00 std.dev.: 0.00,
    Output: 127 rows (6.70kB)
    Output layout: [expr_5, expr_4, month, round, year, count, concat]
    Output partitioning: SINGLE []
Project[]
    | Layout: [expr_5:bigint, expr_4:integer, month:integer, round:double, year:integer, count:bigint, concat:vchar]
    | Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    | CPU: 8.00ms (0.28%), Scheduled: 30.00ms (0.20%), Blocked: 0.00ns (0.00%), Output: 127 rows (6.70kB)
    | Input avg.: 63.50 rows, Input std.dev.: 13.39%
    | expr_5 := BIGINT '2'
    | expr_4 := 2
    | round := round("avg", 2)
    | concat := concat(CAST("year" AS VARCHAR), VARCHAR '-', lpad(CAST("month" AS VARCHAR), BIGINT '2', '0'))
    |
Aggregate[type = FINAL, keys = [year, month]]
    | Layout: [year:integer, month:integer, avg:double, count:bigint]
    | Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
    | CPU: 4.00ms (0.14%), Scheduled: 9.00ms (0.06%), Blocked: 0.00ns (0.00%), Output: 127 rows (3.47kB)
    | Input avg.: 119.00 rows, Input std.dev.: 13.45%
    | avg := avg("avg_8")
    | count := count("count_9")
    |
LocalExchange[partitioning = HASH, arguments = ["year", "month"]]
    | Layout: [year:integer, month:integer, avg_8:row(double, bigint), count_9:bigint]

```

```

    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    |
    CPU: 1.00ms (0.03%), Scheduled: 1.00ms (0.01%), Blocked: 8.53s (12.88%), Output: 238 rows (8.83kB)
    |
    Input avg.: 119.00 rows, Input std.dev.: 0.84%
    |
RemoteSource[sourceFragmentIds = [4]] Layout: [year:integer, month:integer, avg_8:row(double, bigint), count_9:bigint]
    CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 8.54s (12.90%), Output: 238 rows (8.83kB)
    Input avg.: 119.00 rows, Input std.dev.: 0.84%

Fragment 4 [HASH]
    CPU: 1.28s, Scheduled: 7.39s, Blocked 8.86s (Input: 5.94s, Output: 0.00ns), Input: 1538741 rows (35.22MB); per task: avg.: 1538741.00 std.dev.: 0.00,
    Output: 238 rows (9.76kB)
    Output layout: [year, month, avg_8, count_9]
    Output partitioning: HASH [year, month]
    Aggregate[type = PARTIAL, keys = [year, month]]
    |
    Layout: [year:integer, month:integer, avg_8:row(double, bigint), count_9:bigint]
    |
    CPU: 235.00ms (8.11%), Scheduled: 1.16s (7.85%), Blocked: 0.00ns (0.00%), Output: 238 rows (9.76kB)
    |
    Input avg.: 769370.50 rows, Input std.dev.: 0.09%
    |
    avg_8 := avg("expr")
    |
    count_9 := count("product_sk") (mask = product_sk$distinct)
    |
MarkDistinct[distinct = [year:integer, month:integer, product_sk:integer], marker = product_sk$distinct]
    Layout: [product_sk:integer, year:integer, expr:bigint, month:integer, product_sk$distinct:boolean]
    |
    CPU: 740.00ms (25.55%), Scheduled: 4.47s (30.15%), Blocked: 0.00ns (0.00%), Output: 1538741 rows (38.15MB)
    |
    Input avg.: 769370.50 rows, Input std.dev.: 0.09%
    |
LocalExchange[partitioning = HASH, arguments = ["year", "month", "product_sk"]]
    Layout: [product_sk:integer, year:integer, expr:bigint, month:integer]
    |
    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    |
    CPU: 256.00ms (8.84%), Scheduled: 1.56s (10.49%), Blocked: 2.97s (4.48%), Output: 1538741 rows (35.22MB)
    |
    Input avg.: 769370.50 rows, Input std.dev.: 11.68%
    |
RemoteSource[sourceFragmentIds = [5]] Layout: [product_sk:integer, year:integer, expr:bigint, month:integer]
    CPU: 32.00ms (1.10%), Scheduled: 148.00ms (1.00%), Blocked: 5.94s (8.97%), Output: 1538741 rows (35.22MB)
    Input avg.: 769370.50 rows, Input std.dev.: 11.68%

Fragment 5 [HASH]
    CPU: 788.67ms, Scheduled: 3.83s, Blocked 11.58s (Input: 4.06s, Output: 0.00ns), Input: 1542597 rows (22.07MB); per task: avg.: 1542597.00 std.dev.:
    0.00, Output: 1538741 rows (35.22MB)
    Output layout: [product_sk, year, expr, month]
    Output partitioning: HASH [year, month, product_sk]
    Project[]
    |
    Layout: [product_sk:integer, year:integer, expr:bigint, month:integer]
    |
    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    |
    CPU: 336.00ms (11.60%), Scheduled: 1.69s (11.43%), Blocked: 0.00ns (0.00%), Output: 1538741 rows (35.22MB)
    |
    Input avg.: 769370.50 rows, Input std.dev.: 10.34%
    |
    expr := CAST("nutriscore_score" AS bigint)
    |
InnerJoin[criteria = ("date_sk" = "date_sk_0"), distribution = PARTITIONED]
    Layout: [product_sk:integer, nutriscore_score:integer, year:integer, month:integer]
    |
    Estimates: {rows: ? (?), cpu: ?, memory: ?, network: 0B}
    |
    CPU: 430.00ms (14.85%), Scheduled: 1.97s (13.26%), Blocked: 0.00ns (0.00%), Output: 1538741 rows (29.35MB)
    |
    Left (probe)
    |
    Input avg.: 769370.50 rows, Input std.dev.: 10.34%
    |
    Right (build)
    |
    Input avg.: 1928.00 rows, Input std.dev.: 1.92%
    |
    Distribution: PARTITIONED
    |
    dynamicFilterAssignments = {date_sk_0 -> #df_875}
    |
RemoteSource[sourceFragmentIds = [6]]
    Layout: [product_sk:integer, date_sk:integer, nutriscore_score:integer]
    |
    CPU: 21.00ms (0.73%), Scheduled: 133.00ms (0.90%), Blocked: 3.50s (5.29%), Output: 1538741 rows (22.01MB)
    |
    Input avg.: 769370.50 rows, Input std.dev.: 10.34%
    |
LocalExchange[partitioning = HASH, arguments = ["date_sk_0"]]
    Layout: [date_sk_0:integer, year:integer, month:integer]
    |
    Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
    |
    CPU: 0.00ns (0.00%), Scheduled: 8.00ms (0.05%), Blocked: 602.00ms (0.91%), Output: 3856 rows (56.48kB)
    |
    Input avg.: 1928.00 rows, Input std.dev.: 100.00%
    |
RemoteSource[sourceFragmentIds = [7]] Layout: [date_sk_0:integer, year:integer, month:integer]
    CPU: 0.00ns (0.00%), Scheduled: 0.00ns (0.00%), Blocked: 569.00ms (0.86%), Output: 3856 rows (56.48kB)
    Input avg.: 1928.00 rows, Input std.dev.: 100.00%

Fragment 6 [SOURCE]
    CPU: 790.74ms, Scheduled: 3.48s, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 4770925 rows (68.25MB); per task: avg.: 4770925.00 std.dev.:
    0.00, Output: 1538741 rows (22.01MB)

```



```

Output layout: [product_sk, date_sk, nutriscore_score]
Output partitioning: HASH [date_sk]
ScanFilter[table = hive:nutriscan_gold:fact_nutrition, filterPredicate = (NOT ("nutriscore_score" IS NULL)), dynamicFilters = {"date_sk" = #df_875}]
Layout: [product_sk:integer, date_sk:integer, nutriscore_score:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}/{rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 790.00ms (27.28%), Scheduled: 3.47s (23.40%), Blocked: 0.00ns (0.00%), Output: 1538741 rows (22.01MB)
  Input avg.: 681560.71 rows, Input std.dev.: 244.95% product_sk := product_sk:int:REGULAR nutriscore_score := nutriscore_score:int:REGULAR date_sk :=
date_sk:int:REGULAR Input: 4770925 rows (68.25MB), Filtered: 67.75%, Physical input: 26.17MB, Physical input time: 383.95ms
Dynamic filters:
  - df_875, ALL, collection time=512.30ms

Fragment 7 [SOURCE]
  CPU: 13.41ms, Scheduled: 86.36ms, Blocked 0.00ns (Input: 0.00ns, Output: 0.00ns), Input: 3856 rows (56.48kB); per task: avg.: 3856.00 std.dev.: 0.00,
Output: 3856 rows (56.48kB)
Output layout: [date_sk_0, year, month]
Output partitioning: HASH [date_sk_0]
TableScan[table = hive:nutriscan_gold:dim_date] Layout: [date_sk_0:integer, year:integer, month:integer]
  Estimates: {rows: ? (?), cpu: ?, memory: 0B, network: 0B}
  CPU: 13.00ms (0.45%), Scheduled: 86.00ms (0.58%), Blocked: 0.00ns (0.00%), Output: 3856 rows (56.48kB)
  Input avg.: 3856.00 rows, Input std.dev.: 0.00% date_sk_0 := date_sk:int:REGULAR month := month:int:REGULAR year := year:int:REGULAR Input: 3856 rows
(56.48kB), Physical input: 32.92kB, Physical input time: 13.37ms

```

